GRADE 7 COMMON CORE MATH

Answer Key

Main • Backward Branches • Forward 1 (Grade 8) • Forward 2 (Grade 9)

70 question sets • 1755 questions

Page numbers match the Question Collection.

Every page shows its question ID (for example S40-F2-Q3).

CONTENTS

Set 1	**7.RP.A.1**	Compute unit rates associated with ratios of fractions	7
Set 2	**7.RP.A.2.a**	Decide whether two quantities in a table are in a proportional relationship	38
Set 3	**7.RP.A.2.a**	Decide whether a graph shows a proportional relationship	69
Set 4	**7.RP.A.2.b**	Identify the constant of proportionality in tables, graphs, equations, and descriptio	100
Set 5	**7.RP.A.2.c**	Represent a proportional relationship with an equation	131
Set 6	**7.RP.A.2.d**	Explain what a point on the graph of a proportional relationship means	162
Set 7	**7.RP.A.3**	Solve multistep ratio problems	193
Set 8	**7.RP.A.3**	Find a percent increase or a percent decrease	224
Set 9	**7.RP.A.3**	Find a price after a markup or a markdown (discount)	255
Set 10	**7.RP.A.3**	Find simple interest	286
Set 62	**7.RP.A.3**	Find a tip and the total cost with the tip	317
Set 63	**7.RP.A.3**	Find sales tax and the final cost with tax	348
Set 64	**7.RP.A.3**	Find a commission and pay that includes a commission	379
Set 65	**7.RP.A.3**	Find a percent fee and the total with the fee	410
Set 66	**7.RP.A.3**	Solve multistep percent problems that combine a discount with tax or a tip	441
Set 11	**7.RP.A.3**	Find the percent error of an estimate or a measurement	472
Set 12	**7.NS.A.1.a**	Describe situations in which opposite quantities combine to make 0	503

CONTENTS (continued)

Set 13	**7.NS.A.1.b**	Add rational numbers on a number line	534
Set 14	**7.NS.A.1.b**	Use additive inverses: a number plus its opposite is 0	565
Set 15	**7.NS.A.1.b**	Interpret sums of rational numbers in real-world situations	596
Set 16	**7.NS.A.1.c**	Subtract rational numbers by adding the additive inverse	627
Set 17	**7.NS.A.1.c**	Find the distance between two rational numbers as the absolute value of their diff	658
Set 18	**7.NS.A.1.d**	Use properties of operations to add and subtract rational numbers	689
Set 19	**7.NS.A.2.a**	Multiply rational numbers using the rules for signs	726
Set 20	**7.NS.A.2.a**	Interpret products of rational numbers in real-world situations	757
Set 21	**7.NS.A.2.b**	Divide integers and explain the sign of a quotient	788
Set 22	**7.NS.A.2.b**	Interpret quotients of rational numbers in real-world situations	819
Set 23	**7.NS.A.2.c**	Use properties of operations to multiply and divide rational numbers	850
Set 24	**7.NS.A.2.d**	Use long division to write a rational number as a terminating or repeating decimal	881
Set 25	**7.NS.A.3**	Solve real-world problems with the four operations on rational numbers	912
Set 26	**7.EE.A.1**	Add and subtract linear expressions with rational coefficients	943
Set 27	**7.EE.A.1**	Factor and expand linear expressions with rational coefficients	974
Set 28	**7.EE.A.2**	Rewrite an expression to show how the quantities in a problem are related	1005
Set 29	**7.EE.B.3**	Solve multistep problems with positive and negative rational numbers in any form	1036

CONTENTS (continued)

Set 30	**7.EE.B.3**	Use estimation to judge whether an answer is reasonable	1067
Set 31	**7.EE.B.4.a**	Solve word problems that lead to equations of the form px + q = r	1098
Set 32	**7.EE.B.4.a**	Solve word problems that lead to equations of the form p(x + q) = r	1129
Set 33	**7.EE.B.4.a**	Compare an algebraic solution with an arithmetic solution	1160
Set 34	**7.EE.B.4.b**	Solve word problems that lead to inequalities of the form px + q > r or px + q < r	1191
Set 35	**7.EE.B.4.b**	Graph the solution set of an inequality and interpret it in context	1222
Set 36	**7.G.A.1**	Find actual lengths and areas from a scale drawing	1253
Set 37	**7.G.A.1**	Reproduce a scale drawing at a different scale	1284
Set 38	**7.G.A.2**	Draw triangles from given conditions	1315
Set 39	**7.G.A.2**	Decide whether given conditions make one triangle, more than one triangle, or no	1346
Set 40	**7.G.A.3**	Describe the cross sections made by slicing right prisms and pyramids	1377
Set 41	**7.G.B.4**	Find the circumference of a circle	1408
Set 42	**7.G.B.4**	Find the area of a circle	1439
Set 43	**7.G.B.4**	Explain the relationship between the circumference and the area of a circle	1470
Set 44	**7.G.B.5**	Write and solve equations for supplementary and complementary angles	1501
Set 45	**7.G.B.5**	Use vertical and adjacent angles to find unknown angles	1532
Set 46	**7.G.B.6**	Find the area of composite figures made of triangles and quadrilaterals	1563

CONTENTS (continued)

Set 47	**7.G.B.6**	Find the volume of right prisms	1594
Set 48	**7.G.B.6**	Find the surface area of right prisms	1625
Set 67	**7.G.B.6**	Find the volume of a solid made of right prisms	1656
Set 68	**7.G.B.6**	Find the exterior surface area of a solid made of right prisms	1687
Set 49	**7.SP.A.1**	Understand random sampling and representative samples	1718
Set 50	**7.SP.A.2**	Use data from a random sample to draw inferences about a population	1749
Set 51	**7.SP.A.2**	Use several random samples to judge how much estimates vary	1780
Set 69	**7.SP.A.2**	Generate several same-size random samples with random digits and see how the	1811
Set 52	**7.SP.B.3**	Describe how much two distributions overlap using the difference in means as a m	1842
Set 53	**7.SP.B.4**	Compare two populations using measures of center and variability from random s	1873
Set 54	**7.SP.C.5**	Understand probability as a number from 0 to 1 that shows how likely an event is	1904
Set 55	**7.SP.C.6**	Estimate a probability from relative frequency and make predictions	1935
Set 70	**7.SP.C.6**	Collect data from a chance process and use it to estimate a probability	1966
Set 56	**7.SP.C.7.a**	Develop a uniform probability model and use it to find probabilities	1997
Set 57	**7.SP.C.7.b**	Develop a probability model from observed frequencies	2028
Set 58	**7.SP.C.7**	Compare probabilities from a model with observed frequencies and explain any dif	2059
Set 59	**7.SP.C.8.a**	Find the probability of a compound event as the fraction of outcomes in the samp	2090

CONTENTS (continued)

Set 60	**7.SP.C.8.b**	Represent the sample space of a compound event with a list, a table, or a tree dia	2121
Set 61	**7.SP.C.8.c**	Design and use a simulation to estimate the probability of a compound event	2152

SET 1 • 7.RP.A.1 • RATIOS & PROPORTIONAL RELATIONSHIPS

Compute unit rates associated with ratios of fractions

MAIN	Grade 7 main question	**7.RP.A.1**
BACKWARD 1	Find a unit rate from whole-number quantities	**6.RP.A.2**
BACKWARD 2	Divide a fraction by a fraction	**6.NS.A.1**
FORWARD 1	Compare unit rates shown in different representations	**8.EE.B.5**
FORWARD 2	Convert rates from one unit to another	**HSN-Q.A.1**

SET 1 • 7.RP.A.1

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

Ana walks $\frac{3}{4}$ mile in $\frac{1}{3}$ hour at a steady pace.

How many miles does she walk per hour?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q1/5

9

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

2 1/4 miles per hour

GRADING NOTE 9/4 or 2.25 miles per hour is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q1/5

A recipe uses $\frac{2}{3}$ cup of sugar for every $\frac{1}{2}$ cup of flour.

How many cups of sugar are used per cup of flour?

Cups of sugar: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q2/5

10

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

1 1/3 cups

GRADING NOTE 4/3 cups is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q2/5

Leo paints $\frac{5}{6}$ of a wall in $\frac{1}{2}$ hour.

How many walls does he paint per hour?

A. $\frac{3}{5}$ wall

B. $1\frac{2}{3}$ walls

C. $\frac{5}{12}$ wall

D. $2\frac{1}{2}$ walls

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q3/5

11

QUESTION ID

S1-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1 2/3 walls

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q3/5

True or false?

A snail crawls $\frac{1}{8}$ meter in $\frac{1}{4}$ minute. Its speed is $\frac{1}{2}$ meter per minute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q4/5

12

QUESTION ID

S1-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q4/5

True or false?

$\frac{2}{5}$ pound of cheese costs $3. The price is $1.20 per pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q5/5

13

QUESTION ID

S1-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The price is $3 ÷ 2/5 = $7.50 per pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • MAIN • Q5/5

SET 1 • 7.RP.A.1

BACKWARD 1

Find a unit rate from whole-number quantities • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

A car travels 180 miles in 3 hours at a steady speed.

What is its speed in miles per hour?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q1/5

15

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q1/5

6 notebooks cost $9.

What is the cost per notebook?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q2/5

16

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$1.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q2/5

A printer prints 120 pages in 8 minutes.

How many pages does it print per minute?

A. 112

B. 960

C. 15

D. 128

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q3/5

17

QUESTION ID

S1-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q3/5

True or false?

Jo reads 90 pages in 5 days at the same rate each day. That is 18 pages per day.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q4/5

18

QUESTION ID

S1-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q4/5

True or false?

4 pounds of grapes cost $24. The price is $8 per pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q5/5

19

QUESTION ID

S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The price is $6 per pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • 7.RP.A.1

BACKWARD 2

Divide a fraction by a fraction • Grade 6 • The Number System • 6.NS.A.1

Divide.

$\frac{3}{4} \div \frac{1}{2}$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q1/5

21

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3/2 = 1 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q1/5

Divide.

$\frac{5}{6} \div \frac{1}{3}$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q2/5

22

QUESTION ID

S1-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5/2 = 2 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q2/5

Divide.

$\frac{5}{8} \div \frac{1}{4}$

A. $1\frac{1}{4}$

B. $\frac{5}{32}$

C. $\frac{2}{5}$

D. $2\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q3/5

23

QUESTION ID

S1-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{2} \div \frac{1}{4} = 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q4/5

24

QUESTION ID

S1-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q4/5

True or false?

$\frac{2}{3} \div \frac{1}{3} = \frac{2}{9}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q5/5

25

QUESTION ID

S1-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • 7.RP.A.1

FORWARD 1

Compare unit rates shown in different representations • Grade 8 • Expressions & Equations • 8.EE.B.5

Which relationship has the greatest unit rate?

A. y = 4.2x

B. A table with the pairs (2, 8) and (5, 20)

C. A line through (0, 0) and (3, 12.3)

D. y = 3.9x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q1/5

27

QUESTION ID

S1-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. y = 4.2x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q1/5

Runner A: a graph of meters (y) against seconds (x) is a line through (0, 0) and (10, 45).

Runner B: d = 4.8t, where d is meters and t is seconds.

Which runner is faster, and by how many meters per second?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q2/5

28

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Runner B, by 0.3 meter per second

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q2/5

Plan A charges $0.15 per text message. The cost for Plan B is shown by a line through (0, 0) and (40, 5), where x is texts and y is dollars.

Which plan costs less per text?

Plan: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q3/5

29

QUESTION ID

S1-F1-Q3

TYPE

Short answer

CORRECT ANSWER

Plan B

GRADING NOTE Plan B costs $0.125 per text, which is less than $0.15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q3/5

True or false?

The graph of y = 2.5x is steeper than the graph of a proportional relationship that passes through (4, 9).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q4/5

30

QUESTION ID

S1-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q4/5

True or false?

A proportional relationship that passes through (6, 15) has a greater unit rate than y = 2.6x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q5/5

31

QUESTION ID

S1-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 1 • Q5/5

SET 1 • 7.RP.A.1

FORWARD 2

Convert rates from one unit to another • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1

A cheetah runs 66 feet per second.

How many miles per hour is this? (1 mile = 5,280 feet)

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q1/5

33

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q1**

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q1/5

True or false?

Walking $\frac{1}{2}$ mile in $\frac{1}{10}$ hour is a speed of 440 feet per minute. (1 mile = 5,280 feet)

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q2/5

34

QUESTION ID

S1-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q2/5

A pump moves 3 gallons every $\frac{1}{4}$ minute.

How many gallons per hour is this?

Gallons per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q3/5

35

QUESTION ID

S1-F2-Q3

TYPE

Short answer

CORRECT ANSWER

720

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q3/5

A runner covers 400 meters in 80 seconds at a steady pace.

What is the speed in kilometers per hour?

A. 18

B. 300

C. 0.3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q4/5

36

QUESTION ID

S1-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q4/5

True or false?

60 miles per hour is the same speed as 1 mile per second.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q5/5

37

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 60 miles per hour is 1 mile per minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S1-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.1 • Set 1 • FORWARD 2 • Q5/5

Decide whether two quantities in a table are in a proportional relationship

MAIN	Grade 7 main question	**7.RP.A.2.a**
BACKWARD 1	Complete a table of equivalent ratios	**6.RP.A.3.a**
BACKWARD 2	Divide multi-digit decimals	**6.NS.B.3**
FORWARD 1	Decide whether a table shows a linear function	**8.F.A.3**
FORWARD 2	Tell equal differences (linear) from equal factors (exponential) ov...	**HSF-LE.A.1.b**

SET 2 • 7.RP.A.2.a

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The table shows a proportional relationship between x and y.

x	2	5	8
y	7	17.5	28

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q1/5

40

QUESTION ID

S2-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q1/5

Which table shows a proportional relationship?

A. x: 2, 4, 6 and y: 3, 5, 7

B. x: 1, 2, 3 and y: 5, 6, 7

C. x: 3, 6, 9 and y: 12, 24, 36

D. x: 1, 3, 5 and y: 4, 12, 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q2/5

41

QUESTION ID

S2-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. x: 3, 6, 9 and y: 12, 24, 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q2/5

The table shows the cost of renting a bike.

Is the cost proportional to the number of hours? Explain.

Hours	1	2	3
Cost ($)	4	9	14

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q3/5

42

QUESTION ID

S2-M-Q3

TYPE

Short answer

CORRECT ANSWER

No. The ratios 4/1, 9/2, and 14/3 are not equal.

GRADING NOTE Any explanation showing the cost-to-hours ratios are not all equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q3/5

The table shows the gallons of water in a tank after some minutes.

Is the number of gallons proportional to the number of minutes? Use the ratios to explain.

Minutes	2	4	6	8
Gallons	5	10	15	21

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q4/5

43

QUESTION ID

S2-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. The ratios gallons ÷ minutes are 2.5, 2.5, 2.5, and 21 ÷ 8 = 2.625, so they are not all equal.

GRADING NOTE Grade both: "no", and evidence that one ratio (21 to 8) differs from the others.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q4/5

True or false?

The table shows a proportional relationship between x and y.

x	1	2	4
y	3	6	10

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q5/5

44

QUESTION ID

S2-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 2 • MAIN • Q5/5

SET 2 • 7.RP.A.2.a

BACKWARD 1

Complete a table of equivalent ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

The table shows equivalent ratios.

What is the missing value?

Cups of flour	2	4	6
Cups of milk	3	?	9

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q1/5

46

QUESTION ID

S2-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q1/5

The table shows equivalent ratios.

What is the missing value?

Boys	5	10	15
Girls	4	8	?

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q2/5

47

QUESTION ID

S2-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q2/5

Which ratio is equivalent to 3 : 5?

A. 9 : 15

B. 6 : 8

C. 3 : 15

D. 5 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q3/5

48

QUESTION ID

S2-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 9 : 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q3/5

True or false?

2 : 7 and 6 : 21 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q4/5

49

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q4/5

True or false?

4 : 6 and 6 : 8 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q5/5

50

QUESTION ID

S2-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 2 • BACKWARD 1 • Q5/5

SET 2 • 7.RP.A.2.a

BACKWARD 2

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

17.5 ÷ 5

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q1/5

52

QUESTION ID

S2-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q1/5

Divide.

8.4 ÷ 2.4

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q2/5

53

QUESTION ID

S2-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q2/5

Divide.

9.6 ÷ 3

A. 3.2

B. 32

C. 0.32

D. 3.02

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q3/5

54

QUESTION ID

S2-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$28 \div 3.5 = 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q4/5

55

QUESTION ID

S2-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q4/5

True or false?

$4.5 \div 1.5 = 0.3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q5/5

56

QUESTION ID

S2-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • 7.RP.A.2.a

FORWARD 1

Decide whether a table shows a linear function • Grade 8 • Functions • 8.F.A.3

True or false?

The table shows a linear function.

x	0	1	2	3
y	4	9	14	19

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q1/5

58

QUESTION ID

S2-F1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q1/5

Which table shows a linear function that is NOT proportional?

A. x: 0, 1, 2 and y: 1, 2, 4

B. x: 0, 1, 2 and y: 0, 4, 8

C. x: 0, 1, 2 and y: 3, 5, 7

D. x: 0, 1, 2 and y: 0, 1, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q2/5

59

QUESTION ID

S2-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. x: 0, 1, 2 and y: 3, 5, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q2/5

The table shows a linear function.

What is its rate of change?

x	1	2	3	4
y	2	5	8	11

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q3/5

60

QUESTION ID

S2-F1-Q3

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q3/5

Is the function in the table linear? Explain.

x	1	2	3	4
y	1	4	9	16

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q4/5

61

QUESTION ID

S2-F1-Q4

TYPE

Short answer

CORRECT ANSWER

No. y increases by 3, then 5, then 7, so the rate of change is not constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q4/5

True or false?

A table with x-values 0, 1, 2 and y-values 7, 9, 11 shows a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q5/5

62

QUESTION ID

S2-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is linear, but y is not 0 when x is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 2 • FORWARD 1 • Q5/5

SET 2 • 7.RP.A.2.a

FORWARD 2

Tell equal differences (linear) from equal factors (exponential) over equal intervals • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b

Does y grow by equal differences or by equal factors over equal intervals of x? Give evidence from the table.

x	0	1	2	3
y	3	6	12	24

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q1/5

64

QUESTION ID

S2-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Equal factors: each y-value is 2 times the one before. The differences 3, 6, and 12 are not equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q1/5

By how much does y change over each interval of 2 in x? Is the growth linear or exponential?

x	0	2	4	6
y	4	10	16	22

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q2/5

65

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

y increases by 6 over each interval of 2, so the growth is linear (equal differences).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q2/5

Each table has x-values 0, 1, 2, 3.

Which y-values change by equal differences over equal intervals?

A. y: 2, 4, 8, 16

B. y: 1, 3, 9, 27

C. y: 1, 2, 4, 7

D. y: 7, 4, 1, −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q3/5

66

QUESTION ID

S2-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y: 7, 4, 1, −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q3/5

True or false?

A table with x-values 0, 1, 2, 3 and y-values 80, 40, 20, 10 shows y decreasing by equal factors over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q4/5

67

QUESTION ID

S2-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q4/5

True or false?

A table with x-values 0, 1, 2, 3 and y-values 2, 6, 18, 54 shows equal differences over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q5/5

68

QUESTION ID

S2-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The differences are 4, 12, and 36; the y-values have an equal factor of 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S2-F2-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.b • Set 2 • FORWARD 2 • Q5/5

SET 3 • 7.RP.A.2.a • RATIOS & PROPORTIONAL RELATIONSHIPS

Decide whether a graph shows a proportional relationship

MAIN	Grade 7 main question	**7.RP.A.2.a**
BACKWARD 1	Plot the pairs from a ratio table	**6.RP.A.3.a**
BACKWARD 2	Find the coordinates of points in all four quadrants	**6.NS.C.6.c**
FORWARD 1	Recognize linear functions whose graphs do not pass through the…	**8.F.A.3**
FORWARD 2	Interpret the intercepts of a graph in context	**HSF-IF.B.4**

SET 3 • 7.RP.A.2.a

MAIN

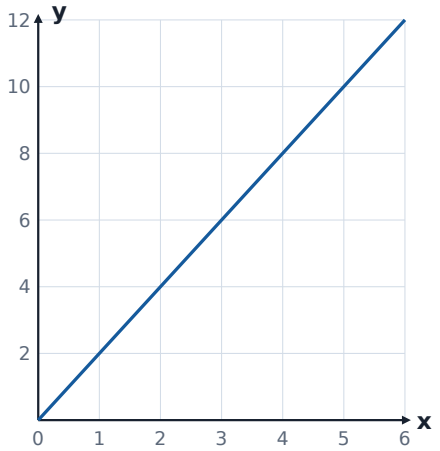
Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The graph shows a proportional relationship between x and y.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q1/5

71

QUESTION ID

S3-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

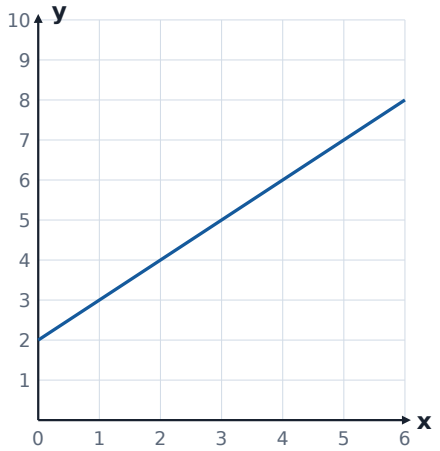
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ID S3-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q1/5

Is the relationship in the graph proportional? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q2/5

72

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. The line does not pass through the origin (0, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q2/5

Which graph shows a proportional relationship?

A. A straight line that crosses the y-axis at 3

B. Points that do not lie on one line

C. A curve through the origin

D. A straight line through the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q3/5

73

QUESTION ID

S3-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A straight line through the origin

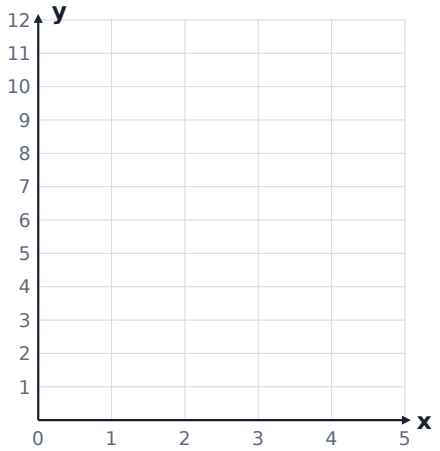
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q3/5

Plot the points (1, 2.5), (2, 5), and (4, 10).

Do the points show a proportional relationship? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q4/5

74

QUESTION ID

S3-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points at (1, 2.5), (2, 5), and (4, 10)

Written: Yes. The points lie on a straight line through the origin; y ÷ x = 2.5 for every point.

GRADING NOTE Grade both: the three points plotted correctly, and "yes" with a reason (same ratio, or a line through the origin).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-M-Q4

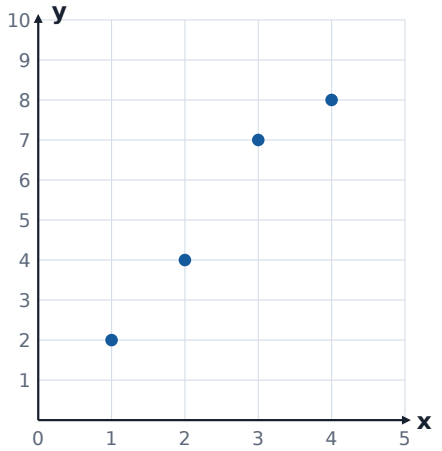
Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q4/5

True or false?

The points in the graph show a proportional relationship.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q5/5

75

QUESTION ID

S3-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 3 • MAIN • Q5/5

SET 3 • 7.RP.A.2.a

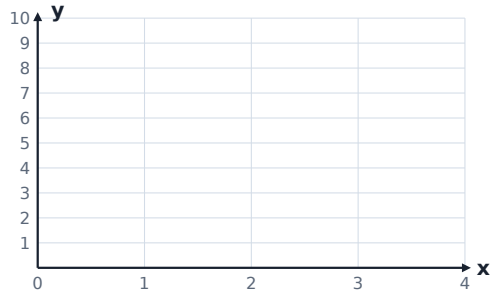
BACKWARD 1

Plot the pairs from a ratio table • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

The table shows equivalent ratios.

Plot each pair (x, y) on the grid.

x	1	2	3
y	3	6	9



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q1/5

77

QUESTION ID

S3-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (1, 3), (2, 6), and (3, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q1/5

A ratio table has the pairs (2, 5), (4, 10), and (6, ?).

What is the missing y-value?

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q2/5

78

QUESTION ID

S3-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q2/5

Which pair belongs in the same ratio table as (3, 4)?

A. (4, 5)

B. (9, 10)

C. (6, 7)

D. (6, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q3/5

79

QUESTION ID

S3-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (6, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q3/5

True or false?

The pairs (2, 6), (4, 12), and (5, 15) can all come from the same ratio table.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q4/5

80

QUESTION ID

S3-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q4/5

True or false?

To plot the pair (3, 7), you start at the origin, move 7 units right, and then move 3 units up.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q5/5

81

QUESTION ID

S3-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B1-Q5

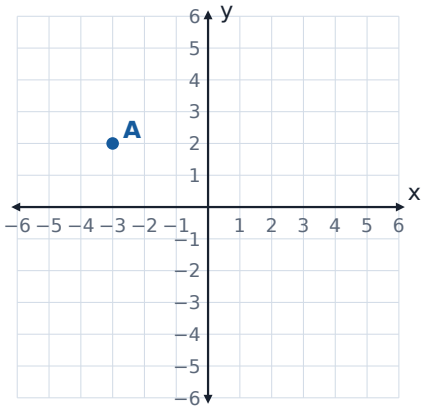
Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 3 • BACKWARD 1 • Q5/5

SET 3 • 7.RP.A.2.a

BACKWARD 2

Find the coordinates of points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

What are the coordinates of point A?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q1/5

83

QUESTION ID

S3-B2-Q1

TYPE

Short answer

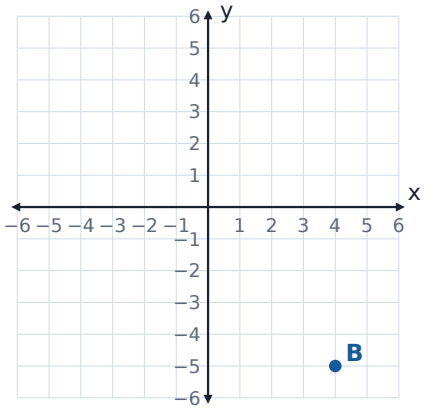
CORRECT ANSWER

(−3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q1/5

What are the coordinates of point B?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q2/5

84

QUESTION ID

S3-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(4, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q2/5

Which point is at the origin?

A. (1, 1)

B. (0, 1)

C. (1, 0)

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q3/5

85

QUESTION ID

S3-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q3/5

True or false?

The point (3, 0) lies on the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q4/5

86

QUESTION ID

S3-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q4/5

True or false?

The point (0, 5) lies on the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q5/5

87

QUESTION ID

S3-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 2 • Q5/5

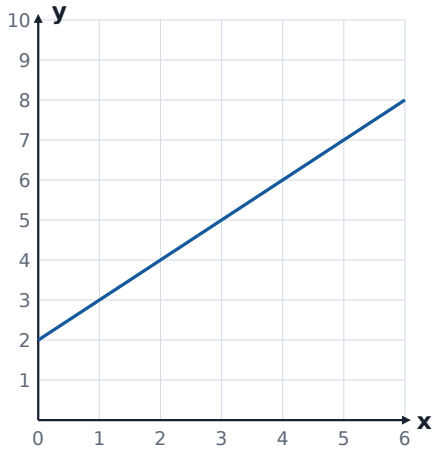
SET 3 • 7.RP.A.2.a

FORWARD 1

Recognize linear functions whose graphs do not pass through the origin • Grade 8 • Functions • 8.F.A.3

The graph shows a line.

Write its equation in the form $y = mx + b$.



Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q1/5

89

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y = x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q1/5

Which equation is linear but NOT proportional?

A. $y = x^2$

B. $y = \frac{3}{x}$

C. $y = 3x$

D. $y = 3x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q2/5

90

QUESTION ID

S3-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. y = 3x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q2/5

True or false?

The graph of y = 4x is a line through the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q3/5

91

QUESTION ID

S3-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q3/5

A line passes through (0, 3) and (2, 7).

Is the relationship proportional? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q4/5

92

QUESTION ID

S3-F1-Q4

TYPE

Short answer

CORRECT ANSWER

No. The line is linear, but it crosses the y-axis at 3, not at the origin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F1-Q4

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q4/5

True or false?

Every linear function is a proportional relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q5/5

93

QUESTION ID

S3-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 8 • Functions • 8.F.A.3 • Set 3 • FORWARD 1 • Q5/5

SET 3 • 7.RP.A.2.a

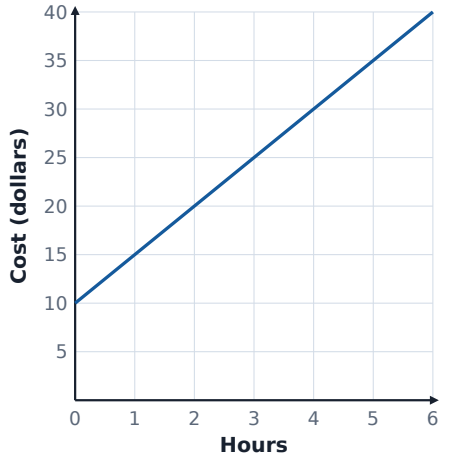
FORWARD 2

Interpret the intercepts of a graph in context • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4

The graph shows the cost of renting a kayak.

What does the point where the line meets the vertical axis mean?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q1/5

95

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

There is a $10 cost at 0 hours (a starting fee).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q1/5

The height of a candle after t hours is h = 24 − 3t centimeters.

After how many hours does the candle burn out (h = 0)?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q2/5

96

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q2/5

The water in a tank after m minutes is w = 50 − 5m gallons.

What does the w-intercept, 50, mean?

A. The tank starts with 50 gallons.

B. The tank drains 50 gallons per minute.

C. The tank is empty after 50 minutes.

D. The tank drains 5 gallons in 50 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q3/5

97

QUESTION ID

S3-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The tank starts with 50 gallons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q3/5

True or false?

The cost of a rental is C = 15 + 2h dollars for h hours. The C-intercept, 15, is the cost when h = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q4/5

98

QUESTION ID

S3-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q4/5

True or false?

For h = 24 − 3t, the h-intercept is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q5/5

99

QUESTION ID

S3-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The h-intercept is 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S3-F2-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 3 • FORWARD 2 • Q5/5

SET 4 • 7.RP.A.2.b • RATIOS & PROPORTIONAL RELATIONSHIPS

Identify the constant of proportionality in tables, graphs, equations, and descriptions

MAIN	Grade 7 main question	**7.RP.A.2.b**
BACKWARD 1	Describe a unit rate	**6.RP.A.2**
BACKWARD 2	Write an equation from a table of values	**6.EE.C.9**
FORWARD 1	Interpret the unit rate as the slope of a graph	**8.EE.B.5**
FORWARD 2	Construct a linear or exponential function from two input-output p…	**HSF-LE.A.2**

SET 4 • 7.RP.A.2.b

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the constant of proportionality (y per x)?

x	2	4	6
y	5	10	15

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q1/5

102

QUESTION ID

S4-M-Q1

TYPE

Short answer

CORRECT ANSWER

2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

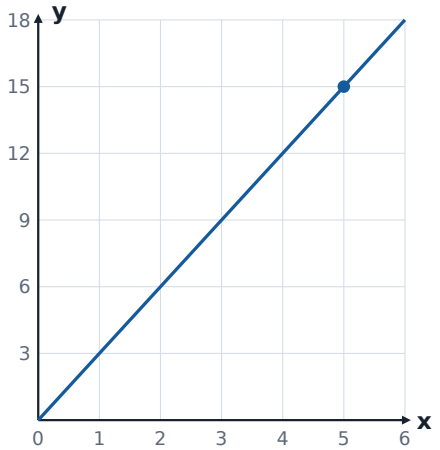
ID S4-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q1/5

The graph shows a proportional relationship.

What is the constant of proportionality?

k = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q2/5

103

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q2/5

What is the constant of proportionality in $y = \frac{2}{3}x$?

A. 2

B. $\frac{3}{2}$

C. 3

D. $\frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q3/5

104

QUESTION ID

S4-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q3/5

True or false?

6 pencils cost $2.10. The constant of proportionality, in dollars per pencil, is 0.35.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q4/5

105

QUESTION ID

S4-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q4/5

Kim types 216 words in 4 minutes at a steady rate.

What is the constant of proportionality in words per minute?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q5/5

106

QUESTION ID

S4-M-Q5

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • MAIN • Q5/5

SET 4 • 7.RP.A.2.b

BACKWARD 1

Describe a unit rate • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

A pack of 8 juice boxes costs $6.

What is the cost per juice box?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q1/5

108

QUESTION ID

S4-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q1/5

A recipe uses 4 cups of flour for 2 loaves.

Which is the unit rate?

A. 6 cups of flour per loaf

B. 8 cups of flour per loaf

C. $\frac{1}{2}$ cup of flour per loaf

D. 2 cups of flour per loaf

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q2/5

109

QUESTION ID

S4-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 2 cups of flour per loaf

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q2/5

Lena swims 600 meters in 12 minutes at a steady pace.

How many meters does she swim per minute?

Meters per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q3/5

110

QUESTION ID

S4-B1-Q3

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q3/5

True or false?

"$3 per pound" is a unit rate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q4/5

111

QUESTION ID

S4-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

5 apples cost $4. The unit rate is $1.25 per apple.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q5/5

112

QUESTION ID

S4-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The unit rate is $0.80 per apple.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • 7.RP.A.2.b

BACKWARD 2

Write an equation from a table of values • Grade 6 • Expressions & Equations • 6.EE.C.9

In the table, y is always 4 times x.

Write an equation for y in terms of x.

x	1	2	3
y	4	8	12

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q1/5

114

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

y = 4x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q1/5

Each ticket costs $7.

Which equation gives the cost c of t tickets?

A. $t = 7c$

B. $c = \frac{t}{7}$

C. $c = 7t$

D. $c = t + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q2/5

115

QUESTION ID

S4-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. c = 7t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q2/5

Write an equation for y in terms of x.

x	2	3	5
y	10	15	25

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q3/5

116

QUESTION ID

S4-B2-Q3

TYPE

Short answer

CORRECT ANSWER

y = 5x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q3/5

True or false?

For y = 6x, when x = 3, y = 18.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q4/5

117

QUESTION ID

S4-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q4/5

True or false?

In d = 50t, d is the independent variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q5/5

118

QUESTION ID

S4-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE t is the independent variable; d depends on t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • 7.RP.A.2.b

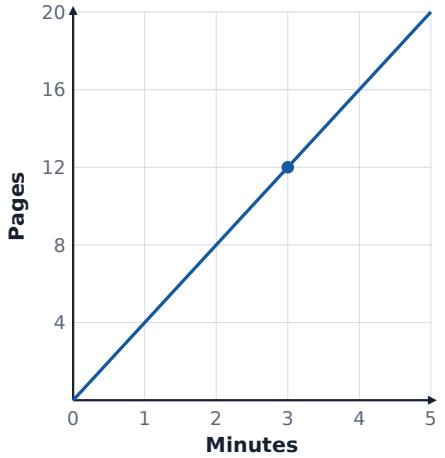
FORWARD 1

Interpret the unit rate as the slope of a graph • Grade 8 • Expressions & Equations • 8.EE.B.5

The graph shows a proportional relationship between minutes and pages read.

What is the slope of the line?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q1/5

120

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A line passes through (0, 0) and (8, 6).

What is its slope?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q2/5

121

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3/4

GRADING NOTE 0.75 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q2/5

The graph of a proportional relationship has a slope of 1.5 liters per minute.

Which point is on the graph?

A. (4, 6)

B. (6, 4)

C. (1.5, 1)

D. (4, 5.5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q3/5

122

QUESTION ID

S4-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (4, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q3/5

True or false?

The slope of the graph of y = 0.8x is 0.8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q4/5

123

QUESTION ID

S4-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q4/5

True or false?

A line through the origin and (2, 9) has a slope of $\frac{2}{9}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q5/5

124

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The slope is 9/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 4 • FORWARD 1 • Q5/5

SET 4 • 7.RP.A.2.b

FORWARD 2

Construct a linear or exponential function from two input-output pairs, a sequence, or a description • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

A linear function has f(2) = 7 and f(6) = 19.

Write f(x).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q1/5

126

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q1/5

An arithmetic sequence is 11, 15, 19, 23, ...

Write a function a(n) for the nth term, starting with n = 1.

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q2/5

127

QUESTION ID

S4-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4n + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q2/5

A tank drains at a constant rate. It holds 500 gallons after 4 minutes and 380 gallons after 10 minutes.

Which function gives the gallons g(t) after t minutes?

A. g(t) = −20t + 500

B. g(t) = 20t + 420

C. g(t) = −20t + 580

D. g(t) = −12t + 548

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q3/5

128

QUESTION ID

S4-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. g(t) = −20t + 580

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q3/5

True or false?

The sequence 2, 6, 18, 54, ... is given by f(n) = 2 · 3^(n − 1) for n = 1, 2, 3, ...

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q4/5

129

QUESTION ID

S4-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q4/5

True or false?

A gym's total cost grows at a constant monthly rate. It is $35 after 1 month and $95 after 5 months. The function is C(m) = 15m + 35.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q5/5

130

QUESTION ID

S4-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The rate is $15 per month, so C(m) = 15m + 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S4-F2-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 4 • FORWARD 2 • Q5/5

Represent a proportional relationship with an equation

MAIN	Grade 7 main question	**7.RP.A.2.c**
BACKWARD 1	Write an equation for a dependent variable	**6.EE.C.9**
BACKWARD 2	Solve unit rate problems	**6.RP.A.3.b**
FORWARD 1	Construct a linear function y = mx + b from a situation	**8.F.B.4**
FORWARD 2	Write an equation in two variables for a situation	**HSA-CED.A.2**

SET 5 • 7.RP.A.2.c

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

ANSWER KEY Reduced copy of the question page, with its answer

Peaches cost $2.40 per pound.

Write an equation for the cost c of p pounds.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q1/5

133

QUESTION ID

S5-M-Q1

TYPE

Short answer

CORRECT ANSWER

c = 2.4p

GRADING NOTE c = 2.40p is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q1/5

A car travels 29 miles per gallon of gas.

Which equation gives the miles m for g gallons?

A. $m = 29g$

B. $m = g + 29$

C. $g = 29m$

D. $m = \frac{g}{29}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q2/5

134

QUESTION ID

S5-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. m = 29g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows a proportional relationship.

Write an equation for y in terms of x.

x	4	8	12
y	3	6	9

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q3/5

135

QUESTION ID

S5-M-Q3

TYPE

Short answer

CORRECT ANSWER

y = 0.75x

GRADING NOTE y = (3/4)x is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q3/5

True or false?

Notebooks cost $1.75 each. The equation c = 1.75n gives the cost c of n notebooks.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q4/5

136

QUESTION ID

S5-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q4/5

True or false?

A printer prints 12 pages every 3 minutes. The equation p = 12m gives the pages p printed in m minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q5/5

137

QUESTION ID

S5-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The equation is p = 4m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 5 • MAIN • Q5/5

SET 5 • 7.RP.A.2.c

BACKWARD 1

Write an equation for a dependent variable • Grade 6 • Expressions & Equations • 6.EE.C.9

A worker earns $18 per hour.

Write an equation for the earnings e after h hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q1/5

139

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

e = 18h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q1/5

Which equation matches the table?

x	1	2	3
y	9	18	27

A. y = x + 8

B. y = 3x

C. x = 9y

D. y = 9x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q2/5

140

QUESTION ID

S5-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. y = 9x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q2/5

Water fills a tank at 6 liters per minute.

Write an equation for the liters L after t minutes.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q3/5

141

QUESTION ID

S5-B1-Q3

TYPE

Short answer

CORRECT ANSWER

L = 6t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q3/5

True or false?

In e = 18h, the value of e depends on the value of h.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q4/5

142

QUESTION ID

S5-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q4/5

True or false?

The point (2, 7) is on the graph of y = 5x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q5/5

143

QUESTION ID

S5-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • 7.RP.A.2.c

BACKWARD 2

Solve unit rate problems • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

5 bags of rice cost $15.

At the same rate, how much do 8 bags cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q1/5

145

QUESTION ID

S5-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q1/5

A machine makes 140 parts in 7 hours.

At the same rate, how many parts does it make in 10 hours?

Parts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q2/5

146

QUESTION ID

S5-B2-Q2

TYPE

Short answer

CORRECT ANSWER

200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q2/5

A bus travels at 45 miles per hour.

How long does it take to travel 135 miles?

A. 3 hours

B. 2 hours

C. 4 hours

D. 6 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q3/5

147

QUESTION ID

S5-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q3/5

True or false?

At $4 per pound, 6 pounds cost $24.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q4/5

148

QUESTION ID

S5-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q4/5

True or false?

A printer prints 12 pages per minute. It takes 6 minutes to print 60 pages.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q5/5

149

QUESTION ID

S5-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It takes 5 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 5 • BACKWARD 2 • Q5/5

SET 5 • 7.RP.A.2.c

FORWARD 1

Construct a linear function y = mx + b from a situation • Grade 8 • Functions • 8.F.B.4

A plumber charges $60 for a visit plus $45 per hour.

Write an equation for the cost y after x hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q1/5

151

QUESTION ID

S5-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y = 45x + 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q1/5

The table shows a linear function.

Write its equation in the form y = mx + b.

x	0	1	2
y	7	10	13

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q2/5

152

QUESTION ID

S5-F1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 3x + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q2/5

A line has a slope of 2 and passes through (0, −5).

Which is its equation?

A. y = −5x + 2

B. y = −2x − 5

C. y = 2x − 5

D. y = 2x + 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q3/5

153

QUESTION ID

S5-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. y = 2x − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q3/5

True or false?

A pool has 300 gallons and fills at 20 gallons per minute. The equation y = 20x + 300 gives the gallons y after x minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q4/5

154

QUESTION ID

S5-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q4/5

True or false?

A candle 15 cm tall burns 2 cm per hour. Its height after x hours is y = 15x − 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q5/5

155

QUESTION ID

S5-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The height is y = −2x + 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 5 • FORWARD 1 • Q5/5

SET 5 • 7.RP.A.2.c

FORWARD 2

Write an equation in two variables for a situation • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2

Pens cost $2 each and markers cost $3 each. Kai spends exactly $30.

Write an equation using p for pens and m for markers.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q1/5

157

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2p + 3m = 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q1/5

Adult tickets cost $12 and child tickets cost $8. A group pays $96.

Write an equation using a for adult tickets and c for child tickets.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q2/5

158

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12a + 8c = 96

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q2/5

A farm has 40 animals. Every animal is a chicken or a goat.

Which equation relates the number of chickens c and goats g?

A. cg = 40

B. c + g = 40

C. 2c + 4g = 40

D. c = 40g

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q3/5

159

QUESTION ID

S5-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. c + g = 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q3/5

True or false?

The equation 5x + 2y = 50 can describe buying x items at $5 each and y items at $2 each for a total of $50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q4/5

160

QUESTION ID

S5-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q4/5

True or false?

p = 6 and m = 4 is a solution of 2p + 3m = 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q5/5

161

QUESTION ID

S5-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S5-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.2 • Set 5 • FORWARD 2 • Q5/5

Explain what a point on the graph of a proportional relationship means

MAIN	Grade 7 main question	**7.RP.A.2.d**
BACKWARD 1	Explain a pair in a table of equivalent ratios	**6.RP.A.3.a**
BACKWARD 2	Read values from the graph of an equation	**6.EE.C.9**
FORWARD 1	Interpret the rate of change and initial value of a linear function	**8.F.B.4**
FORWARD 2	Calculate the average rate of change over an interval	**HSF-IF.B.6**

SET 6 • 7.RP.A.2.d

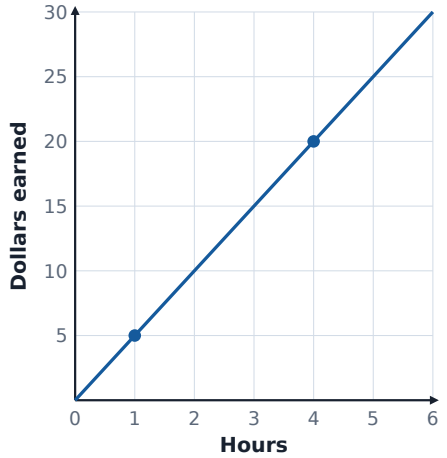
MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph shows the money Raj earns.

What does the point (4, 20) mean in this situation?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q1/5

164

QUESTION ID

S6-M-Q1

TYPE

Short answer

CORRECT ANSWER

Raj earns $20 for working 4 hours.

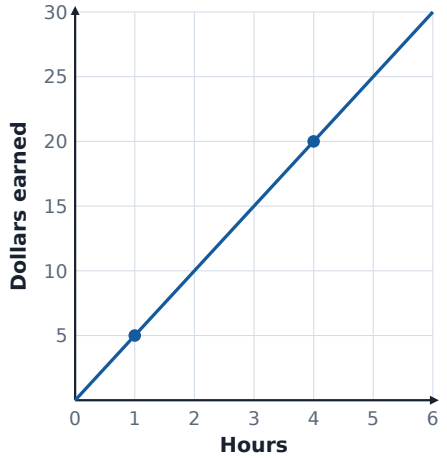
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q1/5

The graph shows the money Raj earns.

What does the point (1, 5) tell you?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q2/5

165

QUESTION ID

S6-M-Q2

TYPE

Short answer

CORRECT ANSWER

Raj earns $5 per hour. This is the unit rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q2/5

The graph shows the money Raj earns.

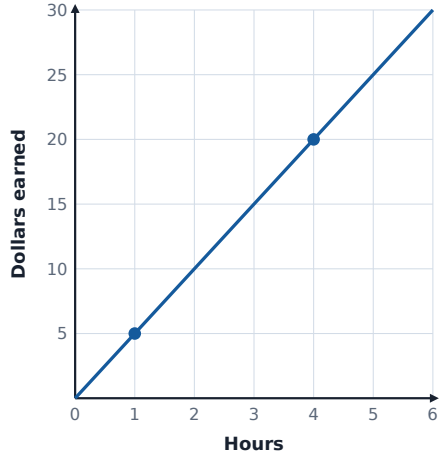
What does the point (0, 0) mean?

A. Raj earns $0 when he works 0 hours.

B. Raj earns $5 per hour.

C. Raj works 0 hours to earn $5.

D. Raj starts with $5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q3/5

166

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Raj earns $0 when he works 0 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q3/5

True or false?

A proportional graph shows cups of rice (x) and cups of water (y). The point (1, 3.5) means 3.5 cups of water are used for each cup of rice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q4/5

167

QUESTION ID

S6-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q4/5

True or false?

A car's distance graph passes through (2, 110), where x is hours and y is miles. This means the car travels 110 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q5/5

168

QUESTION ID

S6-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means 110 miles in 2 hours, or 55 miles per hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 6 • MAIN • Q5/5

SET 6 • 7.RP.A.2.d

BACKWARD 1

Explain a pair in a table of equivalent ratios • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

The table shows the cost of apples.

What does the pair (2, 6) mean?

Pounds	1	2	3
Cost ($)	3	6	9

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q1/5

170

QUESTION ID

S6-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2 pounds of apples cost $6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q1/5

The table shows the time a swimmer takes.

What does the pair (4, 10) mean?

Laps	2	4	6
Minutes	5	10	15

A. Each lap takes 4 minutes.

B. 4 laps take 10 minutes.

C. 10 laps take 4 minutes.

D. Each lap takes 10 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q2/5

171

QUESTION ID

S6-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 4 laps take 10 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q2/5

True or false?

A table shows hours (x) and miles walked (y). The pair (3, 12) means 12 miles are walked in 3 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q3/5

172

QUESTION ID

S6-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q3/5

One ticket costs $8.

What pair (tickets, cost) shows the cost of 5 tickets?

Pair: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q4/5

173

QUESTION ID

S6-B1-Q4

TYPE

Short answer

CORRECT ANSWER

(5, 40)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q4/5

True or false?

A table shows boxes (x) and pencils (y). The pair (2, 24) means each box holds 2 pencils.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q5/5

174

QUESTION ID

S6-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It means 2 boxes hold 24 pencils.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • BACKWARD 1 • Q5/5

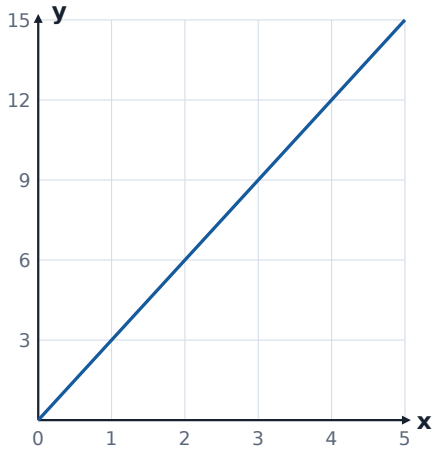
SET 6 • 7.RP.A.2.d

BACKWARD 2

Read values from the graph of an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

The graph shows y = 3x.

What is y when x = 4?



y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q1/5

176

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

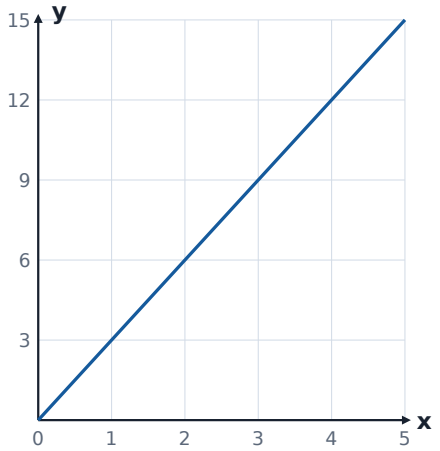
12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q1/5

The graph shows y = 3x.

What is x when y = 6?



x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q2/5

177

QUESTION ID

S6-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q2/5

Which point is on the graph of y = 2x?

A. (6, 3)

B. (4, 6)

C. (3, 6)

D. (2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q3/5

178

QUESTION ID

S6-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q3/5

True or false?

The point (5, 15) is on the graph of y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q4/5

179

QUESTION ID

S6-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q4/5

True or false?

The point (1, 4) is on the graph of y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q5/5

180

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • 7.RP.A.2.d

FORWARD 1

Interpret the rate of change and initial value of a linear function • Grade 8 • Functions • 8.F.B.4

A taxi ride costs y = 2.5x + 4 dollars for x miles.

What does the 4 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q1/5

182

QUESTION ID

S6-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A $4 starting fee (the cost at 0 miles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q1/5

A taxi ride costs y = 2.5x + 4 dollars for x miles.

What does the 2.5 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q2/5

183

QUESTION ID

S6-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The cost increases by $2.50 for each mile.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q2/5

A tank holds y = 200 − 8x gallons after x minutes.

What is the rate of change?

A. −8 gallons per minute

B. 200 gallons per minute

C. 8 gallons

D. −200 gallons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q3/5

184

QUESTION ID

S6-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −8 gallons per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q3/5

True or false?

A plant's height after x weeks is y = 0.5x + 3 centimeters. The plant was 3 cm tall at the start.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q4/5

185

QUESTION ID

S6-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q4/5

True or false?

In y = 12x + 30, the initial value is 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q5/5

186

QUESTION ID

S6-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The initial value is 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 6 • FORWARD 1 • Q5/5

SET 6 • 7.RP.A.2.d

FORWARD 2

Calculate the average rate of change over an interval • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6

$f(x) = x^2$

What is the average rate of change of f from x = 1 to x = 3?

Average rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q1/5

188

QUESTION ID

S6-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q1/5

The table shows the height of a balloon.

What is the average rate of change from 2 seconds to 4 seconds?

Time (s)	0	2	4
Height (m)	0	40	60

Average rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q2/5

189

QUESTION ID

S6-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10 meters per second

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q2/5

A car is 30 miles from home at 1 p.m. and 150 miles from home at 3 p.m.

What is its average rate of change in distance?

A. 75 miles per hour

B. 60 miles per hour

C. 50 miles per hour

D. 120 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q3/5

190

QUESTION ID

S6-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 60 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q3/5

True or false?

For f(x) = 2x + 1, the average rate of change over any interval is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q4/5

191

QUESTION ID

S6-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q4/5

True or false?

For $f(x) = x^2$, the average rate of change from $x = 0$ to $x = 2$ equals the average rate of change from $x = 2$ to $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q5/5

192

QUESTION ID

S6-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are 2 and 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S6-F2-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.6 • Set 6 • FORWARD 2 • Q5/5

Solve multistep ratio problems

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Use ratio language to describe two quantities	**6.RP.A.1**
BACKWARD 2	Use a table of equivalent ratios to solve a problem	**6.RP.A.3.a**
FORWARD 1	Solve linear equations with rational coefficients	**8.EE.C.7.b**
FORWARD 2	Create one-variable equations to solve ratio problems	**HSA-CED.A.1**

SET 7 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Trail mix uses peanuts, raisins, and almonds in a ratio of $5 : 3 : 2$. Kai makes $4\frac{1}{2}$ cups of trail mix.

How many more cups of peanuts than almonds does he use?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q1/5

195

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

1.35 cups

GRADING NOTE 1 7/20 cups or 27/20 cups is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q1/5

A club has boys and girls in a ratio of 4 : 5. There are 36 members.

How many girls are in the club?

Girls: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q2/5

196

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q2/5

Ali mixes blue and yellow paint in a ratio of 2 : 3. He makes 15 liters of paint.

How many liters are blue?

A. 10 liters

B. 9 liters

C. 7.5 liters

D. 6 liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q3/5

197

QUESTION ID

S7-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6 liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q3/5

True or false?

On a map, 2 cm represents 15 km. Two towns are 7 cm apart on the map. A car drives between them at 60 km per hour. The trip takes less than 1 hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q4/5

198

QUESTION ID

S7-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The towns are 52.5 km apart, so the trip takes 52.5 ÷ 60 = 0.875 hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q4/5

True or false?

Juice concentrate and water are mixed in a ratio of 1 : 4. To make 6 liters of juice, you need 6 liters of water.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q5/5

199

QUESTION ID

S7-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE You need 4.8 liters of water.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • MAIN • Q5/5

SET 7 • 7.RP.A.3

BACKWARD 1

Use ratio language to describe two quantities • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1

A class has 12 girls and 15 boys.

Write the ratio of boys to girls.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q1/5

201

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

15 : 12

GRADING NOTE 5 : 4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q1/5

For every 3 red tiles, there are 5 blue tiles.

What is the ratio of blue tiles to all tiles?

A. 5 : 8

B. 5 : 3

C. 8 : 5

D. 3 : 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q2/5

202

QUESTION ID

S7-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 5 : 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q2/5

A team won 9 games and lost 6 games. There were no ties.

Write the ratio of wins to games played.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q3/5

203

QUESTION ID

S7-B1-Q3

TYPE

Short answer

CORRECT ANSWER

9 : 15

GRADING NOTE 3 : 5 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q3/5

True or false?

There are 2 cats for every 7 dogs. The ratio of dogs to cats is 7 : 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q4/5

204

QUESTION ID

S7-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q4/5

True or false?

A bowl has 4 apples and 9 oranges and no other fruit. The ratio of apples to all the fruit is 4 : 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q5/5

205

QUESTION ID

S7-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The ratio is 4 : 13.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • 7.RP.A.3

BACKWARD 2

Use a table of equivalent ratios to solve a problem • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

Lemonade uses 2 cups of mix for every 5 cups of water.

How many cups of water are needed for 8 cups of mix?

Cups of water: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q1/5

207

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q1/5

A store sells 3 notebooks for $4.

At the same rate, how much do 12 notebooks cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q2/5

208

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q2/5

Which pair belongs in a table of ratios equivalent to 3 : 4?

A. 9 : 16

B. 12 : 15

C. 6 : 7

D. 15 : 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q3/5

209

QUESTION ID

S7-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 15 : 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q3/5

True or false?

A table of ratios equivalent to 5 : 2 can include 20 : 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q4/5

210

QUESTION ID

S7-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q4/5

True or false?

A table of ratios equivalent to 2 : 3 can include 8 : 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q5/5

211

QUESTION ID

S7-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 7 • BACKWARD 2 • Q5/5

SET 7 • 7.RP.A.3

FORWARD 1

Solve linear equations with rational coefficients • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for b.

$5b = 3(b + 6)$

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q1/5

213

QUESTION ID

S7-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q1/5

Solve for x.

$\frac{2}{3}x + 4 = 10$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q2/5

214

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q2/5

Solve for x.

$0.4(x - 5) = 2$

A. $x = 0.5$

B. $x = 3$

C. $x = 10$

D. $x = 5.8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q3/5

215

QUESTION ID

S7-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q3/5

True or false?

$x = 12$ is the solution of $\frac{3}{4}x - 2 = 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q4/5

216

QUESTION ID

S7-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q4/5

True or false?

$x = 4$ is the solution of $2(x + 3) = 5x - 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q5/5

217

QUESTION ID

S7-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 7 • FORWARD 1 • Q5/5

SET 7 • 7.RP.A.3

FORWARD 2

Create one-variable equations to solve ratio problems • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1

The ratio of red marbles to green marbles is 3 : 5. There are 14 more green marbles than red marbles.

Write and solve an equation to find the number of red marbles.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q1/5

219

QUESTION ID

S7-F2-Q1

TYPE

Short answer

CORRECT ANSWER

An equation such as 5x − 3x = 14 (x = 7), and the answer 21 red marbles

GRADING NOTE Both parts are required: a valid equation, such as 5x − 3x = 14 or (5/3)r − r = 14 for r red marbles, and the correct answer, 21 red marbles. Any valid equation is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q1/5

A 60-meter rope is cut into two pieces in a ratio of 2 : 3.

Write and solve an equation to find the length of the longer piece.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q2/5

220

QUESTION ID

S7-F2-Q2

TYPE

Short answer

CORRECT ANSWER

An equation such as 2x + 3x = 60 (x = 12), and the answer 36 meters

GRADING NOTE Both parts are required: a valid equation, such as 2x + 3x = 60 or (2/3)L + L = 60 for the longer piece L, and the correct answer, 36 meters. Any valid equation is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q2/5

The ages of two cousins are in a ratio of 4 : 7. The sum of their ages is 66.

Which equation can find x, where the ages are 4x and 7x?

A. $4x \cdot 7x = 66$

B. $4x + 7x = 66$

C. $4 + 7x = 66$

D. $7x - 4x = 66$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q3/5

221

QUESTION ID

S7-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4x + 7x = 66

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q3/5

True or false?

The ratio of a to b is 2 : 5, and a + b = 35. Then a = 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q4/5

222

QUESTION ID

S7-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q4/5

True or false?

The ratio of a to b is 1 : 3, and b − a = 12. Then a = 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q5/5

223

QUESTION ID

S7-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE a = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S7-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 7 • FORWARD 2 • Q5/5

Find a percent increase or a percent decrease

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Find a percent of a quantity	**6.RP.A.3.c**
BACKWARD 2	Find the whole from a part and a percent	**6.RP.A.3.c**
FORWARD 1	Compare quantities written in scientific notation — nearest relate…	**8.EE.A.3**
FORWARD 2	Model a quantity that changes by a constant percent rate	**HSF-LE.A.1.c**

SET 8 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A town's population grew from 2,500 to 3,000.

What is the percent increase?

Percent increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q1/5

226

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

20%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q1/5

The price of a jacket dropped from $80 to $60.

What is the percent decrease?

Percent decrease: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q2/5

227

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q2/5

A plant grew from 40 cm to 46 cm tall.

What is the percent increase?

A. 15%

B. 13%

C. 87%

D. 6%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q3/5

228

QUESTION ID

S8-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 15%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q3/5

True or false?

A score that falls from 50 to 40 is a 20% decrease.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q4/5

229

QUESTION ID

S8-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q4/5

True or false?

A price that rises from $60 to $75 is a 15% increase.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q5/5

230

QUESTION ID

S8-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is a 25% increase.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 8 • MAIN • Q5/5

SET 8 • 7.RP.A.3

BACKWARD 1

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 30% of 70?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q1/5

232

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q1/5

What is 15% of $80?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q2/5

233

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q2/5

What is 4% of 250?

A. 40

B. 100

C. 1

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q3/5

234

QUESTION ID

S8-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q3/5

True or false?

75% of 40 is 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q4/5

235

QUESTION ID

S8-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q4/5

True or false?

20% of 45 is 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q5/5

236

QUESTION ID

S8-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 20% of 45 is 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 1 • Q5/5

SET 8 • 7.RP.A.3

BACKWARD 2

Find the whole from a part and a percent • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

12 is 40% of what number?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q1/5

238

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q1/5

9 students in a class walk to school. That is 25% of the class.

How many students are in the class?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q2/5

239

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q2/5

18 is 60% of what number?

A. 48

B. 12

C. 10.8

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q3/5

240

QUESTION ID

S8-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q3/5

True or false?

15 is 50% of 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q4/5

241

QUESTION ID

S8-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q4/5

True or false?

8 is 20% of 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q5/5

242

QUESTION ID

S8-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 8 is 50% of 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q5/5

SET 8 • 7.RP.A.3

FORWARD 1

Compare quantities written in scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.3

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

City A has about 3×10^6 people. Town B has about 6×10^4 people.

How many times as many people live in City A as in Town B?

Times as many: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q1/5 • NEAREST RELATED

244

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q1/5 • NEAREST RELATED

Write 4,500,000 in scientific notation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q2/5 • NEAREST RELATED

245

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4.5×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q2/5 • NEAREST RELATED

Which number is the greatest?

A. 9×10^6

B. 8×10^5

C. 2×10^7

D. 1×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q3/5 • NEAREST RELATED

246

QUESTION ID

S8-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2×10^7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

8×10^8 is 100 times as large as 8×10^6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q4/5 • NEAREST RELATED

247

QUESTION ID

S8-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

3×10^{-2} is greater than 3×10^{-1}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q5/5 • NEAREST RELATED

248

QUESTION ID

S8-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 8 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 8 • 7.RP.A.3

FORWARD 2

Model a quantity that changes by a constant percent rate • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c

A town of 10,000 people grows by 5% each year.

What is the population after 2 years?

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q1/5

250

QUESTION ID

S8-F2-Q1

TYPE

Short answer

CORRECT ANSWER

11,025

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q1/5

Which situation shows growth by a constant percent rate?

A. A plant grows 2 cm each week.

B. An account earns 3% of its current balance each year.

C. A worker earns $15 more each year.

D. A tank fills 4 gallons each minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q2/5

251

QUESTION ID

S8-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. An account earns 3% of its current balance each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F2-Q2**

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q2/5

A car worth $20,000 loses 10% of its value each year.

What is it worth after 2 years?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q3/5

252

QUESTION ID

S8-F2-Q3

TYPE

Short answer

CORRECT ANSWER

$16,200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q3/5

True or false?

A quantity that is multiplied by 1.08 each year grows by 8% per year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q4/5

253

QUESTION ID

S8-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q4/5

True or false?

A quantity that decreases by 10% each year loses the same number of units each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q5/5

254

QUESTION ID

S8-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S8-F2-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.c • Set 8 • FORWARD 2 • Q5/5

Find a price after a markup or a markdown (discount)

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Multiply decimals	**6.NS.B.3**
BACKWARD 2	Find a percent of an amount of money	**6.RP.A.3.c**
FORWARD 1	Solve percent equations with the variable on both sides	**8.EE.C.7.b**
FORWARD 2	Interpret the parts of an expression in context	**HSA-SSE.A.1.a**

SET 9 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A store pays $25 for a backpack and marks the price up by 40%.

What is the selling price?

Selling price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q1/5

257

QUESTION ID

S9-M-Q1

TYPE

Short answer

CORRECT ANSWER

$35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q1/5

A $120 bike is marked down 25%.

What is the sale price?

Sale price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q2/5

258

QUESTION ID

S9-M-Q2

TYPE

Short answer

CORRECT ANSWER

$90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q2/5

A store buys a lamp for $30 and marks the price up by 60%.

What is the selling price?

A. $36

B. $18

C. $90

D. $48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q3/5

259

QUESTION ID

S9-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q3/5

True or false?

A pair of shoes costs $64. At 15% off, the sale price is $54.40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q4/5

260

QUESTION ID

S9-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q4/5

True or false?

A $50 jacket is marked up 20%. Then the new price is marked down 20%. The final price is $50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q5/5

261

QUESTION ID

S9-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $50 × 1.2 = $60, and $60 × 0.8 = $48.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • MAIN • Q5/5

SET 9 • 7.RP.A.3

BACKWARD 1

Multiply decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

0.15 × 80

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q1/5

263

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q1/5

Multiply.

1.08 × 25

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q2/5

264

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q2/5

Multiply.

0.06×45

A. 270

B. 2.7

C. 0.27

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q3/5

265

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$0.2 \times 36.5 = 7.3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q4/5

266

QUESTION ID

S9-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q4/5

True or false?

$1.5 \times 4.2 = 6.2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q5/5

267

QUESTION ID

S9-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • 7.RP.A.3

BACKWARD 2

Find a percent of an amount of money • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 10% of $56?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q1/5

269

QUESTION ID

S9-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$5.60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q1/5

What is 25% of $48?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q2/5

270

QUESTION ID

S9-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q2/5

What is 8% of $50?

A. $0.40

B. $4

C. $8

D. $40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q3/5

271

QUESTION ID

S9-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q3/5

True or false?

20% of $35 is $7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q4/5

272

QUESTION ID

S9-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q4/5

True or false?

5% of $60 is $30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q5/5

273

QUESTION ID

S9-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 9 • BACKWARD 2 • Q5/5

SET 9 • 7.RP.A.3

FORWARD 1

Solve percent equations with the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Store A sells a game for 20% off its price p plus a $6 shipping fee. Store B sells it for the full price p with free shipping.

Solve 0.8p + 6 = p to find the price at which both stores charge the same.

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q1/5

275

QUESTION ID

S9-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q1/5

Solve for p.

$0.75p + 12 = 0.9p$

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q2/5

276

QUESTION ID

S9-F1-Q2

TYPE

Short answer

CORRECT ANSWER

80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q2/5

Solve for x.

$1.08(x - 5) = 0.08x + 50$

A. x = 45

B. x = 50.6

C. x = 55.4

D. x = 55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q3/5

277

QUESTION ID

S9-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 55.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q3/5

True or false?

p = 40 is the solution of 0.85p + 9 = p + 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q4/5

278

QUESTION ID

S9-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q4/5

True or false?

p = 60 is the solution of 1.1(p + 10) = p + 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q5/5

279

QUESTION ID

S9-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is p = 90.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 9 • FORWARD 1 • Q5/5

SET 9 • 7.RP.A.3

FORWARD 2

Interpret the parts of an expression in context • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A price p is discounted 10%. Then 7% sales tax is added. The final cost is 1.07(0.9p).

What does 0.9p represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q1/5

281

QUESTION ID

S9-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The price after the 10% discount (90% of p).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q1/5

The final cost of an item is 1.07(0.9p).

What does the factor 1.07 represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q2/5

282

QUESTION ID

S9-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Adding 7% sales tax (paying 107% of the discounted price).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q2/5

A store's selling price is p + 0.4p, where p is what the store paid.

What does 0.4p represent?

A. The selling price

B. The amount of a 40% discount

C. The price the store paid

D. The amount of the 40% markup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q3/5

283

QUESTION ID

S9-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The amount of the 40% markup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q3/5

True or false?

The expression 0.85p gives the price after a 15% discount on price p.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q4/5

284

QUESTION ID

S9-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q4/5

True or false?

The expression 1.2p gives the price after a 20% discount on price p.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q5/5

285

QUESTION ID

S9-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1.2p is a 20% increase.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S9-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 9 • FORWARD 2 • Q5/5

Find simple interest

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Evaluate a formula	**6.EE.A.2.c**
BACKWARD 2	Write a percent as a rate per 100 and as a decimal	**6.RP.A.3.c**
FORWARD 1	Model a growing balance with a linear function	**8.F.B.4**
FORWARD 2	Compare simple interest with compound interest	**HSF-LE.A.1.a**

SET 10 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Ella deposits $800 in an account that earns 3% simple interest per year.

How much interest does she earn in 4 years?

Interest: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q1/5

288

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

$96

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q1/5

Leo borrows $2,400 at 5% simple interest per year.

How much interest does he owe after 3 years?

Interest: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q2/5

289

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

$360

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q2/5

An account earns 4% simple interest per year.

How many years does it take $500 to earn $100 in interest?

A. 4 years

B. 5 years

C. 2 years

D. 20 years

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q3/5

290

QUESTION ID

S10-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5 years

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q3/5

True or false?

$1,500 borrowed at 5% simple interest per year for 2 years has $150 in interest.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q4/5

291

QUESTION ID

S10-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q4/5

True or false?

$600 earns 2.5% simple interest per year. The balance after 3 years is $660.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q5/5

292

QUESTION ID

S10-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The interest is $45, so the balance is $645.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • MAIN • Q5/5

SET 10 • 7.RP.A.3

BACKWARD 1

Evaluate a formula • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Use I = prt.

Find I when p = 200, r = 0.05, and t = 3.

I = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q1/5

294

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q1/5

Use A = lw.

Find A when l = 12 and w = 7.5.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q2/5

295

QUESTION ID

S10-B1-Q2

TYPE

Short answer

CORRECT ANSWER

90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q2/5

Use d = rt.

Find d when r = 55 and t = 4.

A. 13.75

B. 220

C. 51

D. 59

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q3/5

296

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 220

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q3/5

True or false?

For P = 2l + 2w, when l = 6 and w = 4, P = 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q4/5

297

QUESTION ID

S10-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q4/5

True or false?

For C = 5n + 3, when n = 4, C = 32.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q5/5

298

QUESTION ID

S10-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE C = 23.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 10 • BACKWARD 1 • Q5/5

SET 10 • 7.RP.A.3

BACKWARD 2

Write a percent as a rate per 100 and as a decimal • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write 6% as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q1/5

300

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.06

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q1/5

Write 2.5% as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q2/5

301

QUESTION ID

S10-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.025

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q2/5

Which decimal is equal to 45%?

A. 4.5

B. 0.45

C. 45

D. 0.045

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q3/5

302

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q3/5

True or false?

$3\% = \frac{3}{100} = 0.03$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q4/5

303

QUESTION ID

S10-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q4/5

True or false?

8% = 0.8

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q5/5

304

QUESTION ID

S10-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • BACKWARD 2 • Q5/5

SET 10 • 7.RP.A.3

FORWARD 1

Model a growing balance with a linear function • Grade 8 • Functions • 8.F.B.4

An account starts with $500 and earns $20 in simple interest each year.

Write a function for the balance B after t years.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q1/5

306

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

20t + 500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q1/5

A $1,000 loan charges $45 in simple interest each year.

Write a function for the amount A owed after t years.

A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q2/5

307

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

45t + 1,000

GRADING NOTE A = 45t + 1000 without the comma is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q2/5

A loan balance after t years is B = 1,200 + 36t.

What is the rate of change?

A. $36 per year

B. $1,236 per year

C. $1,200 per year

D. 3% per year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q3/5

308

QUESTION ID

S10-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $36 per year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q3/5

True or false?

For B = 20t + 500, the balance after 3 years is $560.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q4/5

309

QUESTION ID

S10-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q4/5

True or false?

For A = 45t + 1,000, the initial value is 45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q5/5

310

QUESTION ID

S10-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The initial value is 1,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 1 • Q5/5

SET 10 • 7.RP.A.3

FORWARD 2

Compare simple interest with compound interest • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

$1,000 earns 5% simple interest per year.

What is the balance after 2 years?

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q1/5

312

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$1,100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q1/5

$1,000 earns 5% interest compounded once each year.

What is the balance after 2 years?

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q2/5

313

QUESTION ID

S10-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$1,102.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q2/5

Which kind of interest makes a balance grow by equal differences over equal time intervals?

A. Both kinds

B. Compound interest

C. Simple interest

D. Neither kind

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q3/5

314

QUESTION ID

S10-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Simple interest

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q3/5

True or false?

With compound interest, a balance grows by equal factors over equal time intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q4/5

315

QUESTION ID

S10-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q4/5

True or false?

After 1 year, $500 at 4% compound interest (compounded yearly) earns more than $500 at 4% simple interest.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q5/5

316

QUESTION ID

S10-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE After 1 year, both earn $20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S10-F2-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 10 • FORWARD 2 • Q5/5

SET 62 • 7.RP.A.3 • RATIOS & PROPORTIONAL RELATIONSHIPS

Find a tip and the total cost with the tip

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Find 5%, 10%, and 20% of an amount of money	**6.RP.A.3.c**
BACKWARD 2	Add amounts of money	**6.NS.B.3**
FORWARD 1	Solve tip equations with the variable on both sides	**8.EE.C.7.b**
FORWARD 2	Write a function for a total that combines a tip and a fixed fee	**HSF-BF.A.1**

SET 62 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A meal costs $45. Kai leaves a 20% tip.

What is the total cost of the meal with the tip?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • MAIN • Q1/5

319

QUESTION ID

S62-M-Q1

TYPE

Short answer

CORRECT ANSWER

$54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • MAIN • Q1/5

A haircut costs $28. Mia leaves a 15% tip.

How much is the tip?

Tip: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • MAIN • Q2/5

320

QUESTION ID

S62-M-Q2

TYPE

Short answer

CORRECT ANSWER

$4.20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • MAIN • Q2/5

A restaurant bill is $64. A group adds an 18% tip.

What is the total?

A. $66.88

B. $82

C. $11.52

D. $75.52

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • MAIN • Q3/5

321

QUESTION ID

S62-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $75.52

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • MAIN • Q3/5

True or false?

A 20% tip on a $36.50 meal is $7.30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • MAIN • Q4/5

322

QUESTION ID

S62-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • MAIN • Q4/5

True or false?

A 15% tip on a $40 meal makes the total $55.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • MAIN • Q5/5

323

QUESTION ID

S62-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The tip is $6, so the total is $46.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 62 • MAIN • Q5/5

SET 62 • 7.RP.A.3

BACKWARD 1

Find 5%, 10%, and 20% of an amount of money • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 10% of $42?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 62 • BACKWARD 1 • Q1/5

325

QUESTION ID

S62-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$4.20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 62 • BACKWARD 1 • Q1/5

What is 5% of $42?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 62 • BACKWARD 1 • Q2/5

326

QUESTION ID

S62-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$2.10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 62 • BACKWARD 1 • Q2/5

What is 20% of $55?

A. $5.50

B. $1.10

C. $11

D. $20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 62 • BACKWARD 1 • Q3/5

327

QUESTION ID

S62-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 62 • BACKWARD 1 • Q3/5

True or false?

15% of $20 is $3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 62 • BACKWARD 1 • Q4/5

328

QUESTION ID

S62-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 62 • BACKWARD 1 • Q4/5

True or false?

20% of $30 is $3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 62 • BACKWARD 1 • Q5/5

329

QUESTION ID

S62-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 20% of $30 is $6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 62 • BACKWARD 1 • Q5/5

SET 62 • 7.RP.A.3

BACKWARD 2

Add amounts of money • Grade 6 • The Number System • 6.NS.B.3

Add.

$45.80 + $9.16

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • BACKWARD 2 • Q1/5

331

QUESTION ID

S62-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$54.96

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • BACKWARD 2 • Q1/5

Add.

$23.45 + $4.69

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • BACKWARD 2 • Q2/5

332

QUESTION ID

S62-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$28.14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • BACKWARD 2 • Q2/5

Add.

$18.75 + $3.75

A. $15.00

B. $22.40

C. $22.50

D. $21.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • BACKWARD 2 • Q3/5

333

QUESTION ID

S62-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $22.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • BACKWARD 2 • Q3/5

True or false?

$9.99 + $1.50 = $11.49

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • BACKWARD 2 • Q4/5

334

QUESTION ID

S62-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • BACKWARD 2 • Q4/5

True or false?

$12.60 + $2.52 = $14.12

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • BACKWARD 2 • Q5/5

335

QUESTION ID

S62-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • 7.RP.A.3

FORWARD 1

Solve tip equations with the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Two friends split a meal that costs m dollars. Ana pays half of m plus a $5 tip. Ben pays 40% of m plus a $9 tip. They pay the same amount.

Solve 0.5m + 5 = 0.4m + 9.

m = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 62 • FORWARD 1 • Q1/5

337

QUESTION ID

S62-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 62 • FORWARD 1 • Q1/5

Solve for m.

$1.2(m + 10) = 1.15m + 14.5$

m = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 62 • FORWARD 1 • Q2/5

338

QUESTION ID

S62-F1-Q2

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 62 • FORWARD 1 • Q2/5

Solve for x.

$0.15x + 0.85x = 2(x - 20)$

A. $x = -40$

B. $x = 40$

C. $x = 80$

D. $x = 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 62 • FORWARD 1 • Q3/5

339

QUESTION ID

S62-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 62 • FORWARD 1 • Q3/5

True or false?

$x = 30$ is the solution of $1.2x - 6 = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 62 • FORWARD 1 • Q4/5

340

QUESTION ID

S62-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 62 • FORWARD 1 • Q4/5

True or false?

$x = 25$ is the solution of $0.18x + 2 = 0.1x + 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 62 • FORWARD 1 • Q5/5

341

QUESTION ID

S62-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 50$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 62 • FORWARD 1 • Q5/5

SET 62 • 7.RP.A.3

FORWARD 2

Write a function for a total that combines a tip and a fixed fee • High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1

A delivery app adds a 15% tip and a $4 fee to the price m of a meal.

Write a function T(m) for the total.

T(m) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 62 • FORWARD 2 • Q1/5

343

QUESTION ID

S62-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.15m + 4

GRADING NOTE m + 0.15m + 4 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 62 • FORWARD 2 • Q1/5

A delivery total is T(m) = 1.15m + 4, where m is the meal price in dollars.

Find T(30).

T(30) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 62 • FORWARD 2 • Q2/5

344

QUESTION ID

S62-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$38.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 62 • FORWARD 2 • Q2/5

A 20% tip is added to a meal price m. Then 8% tax is added to that amount.

Which function gives the total?

A. T(m) = 0.2m + 0.08m

B. T(m) = m + 1.2 + 0.08

C. T(m) = 1.28m

D. T(m) = 1.08(1.2m)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 62 • FORWARD 2 • Q3/5

345

QUESTION ID

S62-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. T(m) = 1.08(1.2m)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 62 • FORWARD 2 • Q3/5

True or false?

Adding a fixed $5 fee to a meal price m with a 20% tip gives the function T(m) = 1.2m + 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 62 • FORWARD 2 • Q4/5

346

QUESTION ID

S62-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 62 • FORWARD 2 • Q4/5

True or false?

For T(m) = 1.15m + 4, doubling the meal price doubles the total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 62 • FORWARD 2 • Q5/5

347

QUESTION ID

S62-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The $4 fee does not double: T(20) = 27, but T(40) = 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 62 • FORWARD 2 • Q5/5

Find sales tax and the final cost with tax

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Find a percent of an amount of money	**6.RP.A.3.c**
BACKWARD 2	Multiply by a decimal	**6.NS.B.3**
FORWARD 1	Compare tax and flat charges with equations that have the variab…	**8.EE.C.7.b**
FORWARD 2	Interpret a tax expression by seeing its parts as single quantities	**HSA-SSE.A.1.b**

SET 63 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A jacket costs $80. The sales tax is 6%.

How much is the tax?

Tax: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 63 • MAIN • Q1/5

350

QUESTION ID

S63-M-Q1

TYPE

Short answer

CORRECT ANSWER

$4.80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 63 • MAIN • Q1/5

A book costs $24.50. The sales tax is 8%.

What is the total cost?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 63 • MAIN • Q2/5

351

QUESTION ID

S63-M-Q2

TYPE

Short answer

CORRECT ANSWER

$26.46

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 63 • MAIN • Q2/5

A TV costs $350. The sales tax is 7.5%.

What is the final cost?

A. $376.25

B. $357.50

C. $425

D. $26.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 63 • MAIN • Q3/5

352

QUESTION ID

S63-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $376.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 63 • MAIN • Q3/5

True or false?

A $12 item with 5% sales tax costs $12.60 in all.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 63 • MAIN • Q4/5

353

QUESTION ID

S63-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 63 • MAIN • Q4/5

True or false?

A shirt costs $35. With 6% sales tax, the total is $37.20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 63 • MAIN • Q5/5

354

QUESTION ID

S63-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The total is $37.10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 63 • MAIN • Q5/5

SET 63 • 7.RP.A.3

BACKWARD 1

Find a percent of an amount of money • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 6% of $45?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 63 • BACKWARD 1 • Q1/5

356

QUESTION ID

S63-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$2.70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 63 • BACKWARD 1 • Q1/5

What is 8% of $15?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 63 • BACKWARD 1 • Q2/5

357

QUESTION ID

S63-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$1.20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 63 • BACKWARD 1 • Q2/5

What is 5% of $64?

A. $6.40

B. $32

C. $3.20

D. $0.32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 63 • BACKWARD 1 • Q3/5

358

QUESTION ID

S63-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $3.20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 63 • BACKWARD 1 • Q3/5

True or false?

7% of $200 is $14.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 63 • BACKWARD 1 • Q4/5

359

QUESTION ID

S63-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 63 • BACKWARD 1 • Q4/5

True or false?

4% of $75 is $30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 63 • BACKWARD 1 • Q5/5

360

QUESTION ID

S63-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4% of $75 is $3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 63 • BACKWARD 1 • Q5/5

SET 63 • 7.RP.A.3

BACKWARD 2

Multiply by a decimal • Grade 6 • The Number System • 6.NS.B.3

Multiply.

1.06 × 50

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • BACKWARD 2 • Q1/5

362

QUESTION ID

S63-B2-Q1

TYPE

Short answer

CORRECT ANSWER

53

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

0.08 × 32.5

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • BACKWARD 2 • Q2/5

363

QUESTION ID

S63-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • BACKWARD 2 • Q2/5

Multiply.

1.05×18

A. 19.8

B. 18.9

C. 18.05

D. 1.89

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • BACKWARD 2 • Q3/5

364

QUESTION ID

S63-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 18.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • BACKWARD 2 • Q3/5

True or false?

$0.07 \times 300 = 21$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • BACKWARD 2 • Q4/5

365

QUESTION ID

S63-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • BACKWARD 2 • Q4/5

True or false?

$1.08 \times 25 = 25.08$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • BACKWARD 2 • Q5/5

366

QUESTION ID

S63-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • 7.RP.A.3

FORWARD 1

Compare tax and flat charges with equations that have the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Store A charges the price p plus 8% tax. Store B charges the price p plus a flat $6 with no tax.

Solve 1.08p = p + 6 to find when they charge the same.

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 63 • FORWARD 1 • Q1/5

368

QUESTION ID

S63-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 63 • FORWARD 1 • Q1/5

Solve for p.

$1.05(p + 20) = p + 24$

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 63 • FORWARD 1 • Q2/5

369

QUESTION ID

S63-F1-Q2

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 63 • FORWARD 1 • Q2/5

Solve for x.

$1.1x - 4 = 0.9x + 6$

A. x = 5

B. x = 20

C. x = 10

D. x = 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 63 • FORWARD 1 • Q3/5

370

QUESTION ID

S63-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 63 • FORWARD 1 • Q3/5

True or false?

p = 100 is the solution of 1.06p = p + 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 63 • FORWARD 1 • Q4/5

371

QUESTION ID

S63-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 63 • FORWARD 1 • Q4/5

True or false?

p = 40 is the solution of 1.07p − 2 = p + 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 63 • FORWARD 1 • Q5/5

372

QUESTION ID

S63-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 1.07(40) − 2 = 40.8, but 40 + 1 = 41.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 63 • FORWARD 1 • Q5/5

SET 63 • 7.RP.A.3

FORWARD 2

Interpret a tax expression by seeing its parts as single quantities • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

n items each cost p dollars. With 6% sales tax, the total is 1.06(np).

What does np represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 63 • FORWARD 2 • Q1/5

374

QUESTION ID

S63-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The cost of the items before tax.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F2-Q1**

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 63 • FORWARD 2 • Q1/5

A total with tax is (1 + r)p, where r is the tax rate as a decimal and p is the price.

Find the total when r = 0.075 and p = 40.

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 63 • FORWARD 2 • Q2/5

375

QUESTION ID

S63-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$43

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 63 • FORWARD 2 • Q2/5

An order costs p dollars plus s dollars of shipping, and 8% tax is charged on both. The total is 1.08(p + s).

Which part is the amount being taxed?

A. p + s

B. 1.08

C. s

D. p

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 63 • FORWARD 2 • Q3/5

376

QUESTION ID

S63-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. p + s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 63 • FORWARD 2 • Q3/5

True or false?

1.06p and p + 0.06p are equivalent expressions for a price p with 6% tax.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 63 • FORWARD 2 • Q4/5

377

QUESTION ID

S63-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 63 • FORWARD 2 • Q4/5

True or false?

In 1.08(p + 5), the tax is charged only on p.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 63 • FORWARD 2 • Q5/5

378

QUESTION ID

S63-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The tax is charged on p + 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 63 • FORWARD 2 • Q5/5

SET 64 • 7.RP.A.3 • RATIOS & PROPORTIONAL RELATIONSHIPS

Find a commission and pay that includes a commission

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Find a percent of a quantity	**6.RP.A.3.c**
BACKWARD 2	Find what percent a part is of a whole	**6.RP.A.3.c**
FORWARD 1	Model pay with a base salary plus a commission	**8.F.B.4**
FORWARD 2	Write and solve an inequality to meet an earnings goal	**HSA-CED.A.3**

SET 64 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A salesperson earns a 6% commission. She sells $2,450 of products.

What is her commission?

Commission: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 64 • MAIN • Q1/5

381

QUESTION ID

S64-M-Q1

TYPE

Short answer

CORRECT ANSWER

$147

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 64 • MAIN • Q1/5

A real estate agent earns a 3% commission on a $280,000 home sale.

What is the commission?

Commission: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 64 • MAIN • Q2/5

382

QUESTION ID

S64-M-Q2

TYPE

Short answer

CORRECT ANSWER

$8,400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 64 • MAIN • Q2/5

Jo earns $400 per week plus a 5% commission on her sales. She sells $3,600 in one week.

What is her pay for the week?

A. $4,000

B. $580

C. $180

D. $418

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 64 • MAIN • Q3/5

383

QUESTION ID

S64-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $580

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 64 • MAIN • Q3/5

True or false?

A 12% commission on $750 in sales is $90.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 64 • MAIN • Q4/5

384

QUESTION ID

S64-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 64 • MAIN • Q4/5

True or false?

An agent earns a $240 commission on $4,000 in sales. The commission rate is 4%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 64 • MAIN • Q5/5

385

QUESTION ID

S64-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The rate is 240 ÷ 4,000 = 6%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 64 • MAIN • Q5/5

SET 64 • 7.RP.A.3

BACKWARD 1

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 3% of 2,000?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 1 • Q1/5

387

QUESTION ID

S64-B1-Q1

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 1 • Q1/5

What is 12% of 150?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 1 • Q2/5

388

QUESTION ID

S64-B1-Q2

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 1 • Q2/5

What is 5% of 3,600?

A. 720

B. 18

C. 1,800

D. 180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 1 • Q3/5

389

QUESTION ID

S64-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 1 • Q3/5

True or false?

8% of 500 is 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 1 • Q4/5

390

QUESTION ID

S64-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 1 • Q4/5

True or false?

2% of 900 is 180.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 1 • Q5/5

391

QUESTION ID

S64-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2% of 900 is 18.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 1 • Q5/5

SET 64 • 7.RP.A.3

BACKWARD 2

Find what percent a part is of a whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

240 is what percent of 4,000?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 2 • Q1/5

393

QUESTION ID

S64-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 2 • Q1/5

What percent of 800 is 32?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 2 • Q2/5

394

QUESTION ID

S64-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 2 • Q2/5

45 out of 300 is what percent?

A. 30%

B. 6.67%

C. 15%

D. 45%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 2 • Q3/5

395

QUESTION ID

S64-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 15%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 2 • Q3/5

True or false?

90 is 12% of 750.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 2 • Q4/5

396

QUESTION ID

S64-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 2 • Q4/5

True or false?

25 is 5% of 250.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 2 • Q5/5

397

QUESTION ID

S64-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 25 is 10% of 250.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 64 • BACKWARD 2 • Q5/5

SET 64 • 7.RP.A.3

FORWARD 1

Model pay with a base salary plus a commission • Grade 8 • Functions • 8.F.B.4

A job pays $350 per week plus 4% of sales s.

Write a function for the weekly pay P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 64 • FORWARD 1 • Q1/5

399

QUESTION ID

S64-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.04s + 350

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 64 • FORWARD 1 • Q1/5

Job A pays $500 per week plus 2% of sales. Job B pays $300 per week plus 6% of sales.

For what weekly sales do the two jobs pay the same?

Sales: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 64 • FORWARD 1 • Q2/5

400

QUESTION ID

S64-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$5,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 64 • FORWARD 1 • Q2/5

A line through (0, 200) and (10,000, 900) models weekly pay (y) against sales (x).

What is the commission rate?

A. 7%

B. 9%

C. 70%

D. 0.07%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 64 • FORWARD 1 • Q3/5

401

QUESTION ID

S64-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 7%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 64 • FORWARD 1 • Q3/5

True or false?

For P = 0.05s + 250, the pay with no sales is $250.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 64 • FORWARD 1 • Q4/5

402

QUESTION ID

S64-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 64 • FORWARD 1 • Q4/5

True or false?

For P = 0.03s + 400, each $100 in sales adds $30 in pay.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 64 • FORWARD 1 • Q5/5

403

QUESTION ID

S64-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each $100 adds $3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 64 • FORWARD 1 • Q5/5

SET 64 • 7.RP.A.3

FORWARD 2

Write and solve an inequality to meet an earnings goal • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3

Pay is $300 plus 5% of sales s.

Write an inequality for earning at least $800 in a week.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 64 • FORWARD 2 • Q1/5

405

QUESTION ID

S64-F2-Q1

TYPE

Short answer

CORRECT ANSWER

300 + 0.05s ≥ 800

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 64 • FORWARD 2 • Q1/5

Solve 300 + 0.05s ≥ 800, where s is weekly sales in dollars. Explain what the solution means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 64 • FORWARD 2 • Q2/5

406

QUESTION ID

S64-F2-Q2

TYPE

Short answer

CORRECT ANSWER

s ≥ 10,000; weekly sales must be at least $10,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 64 • FORWARD 2 • Q2/5

Pay is 250 + 0.08s dollars for sales of s dollars.

Which amount of sales meets the goal
250 + 0.08s ≥ 650?

A. $4,500

B. $5,500

C. $4,000

D. $3,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 64 • FORWARD 2 • Q3/5

407

QUESTION ID

S64-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $5,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 64 • FORWARD 2 • Q3/5

True or false?

With pay of 200 + 0.1s dollars, sales of $6,000 meet a goal of at least $750.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 64 • FORWARD 2 • Q4/5

408

QUESTION ID

S64-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 64 • FORWARD 2 • Q4/5

True or false?

With pay of 400 + 0.04s dollars, sales of $5,000 meet a goal of at least $650.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 64 • FORWARD 2 • Q5/5

409

QUESTION ID

S64-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The pay is $600.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.3 • Set 64 • FORWARD 2 • Q5/5

SET 65 • 7.RP.A.3 • RATIOS & PROPORTIONAL RELATIONSHIPS

Find a percent fee and the total with the fee

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Write a percent as a decimal	**6.RP.A.3.c**
BACKWARD 2	Multiply decimals	**6.NS.B.3**
FORWARD 1	Compare fee plans with equations that have the variable on both s…	**8.EE.C.7.b**
FORWARD 2	Write a function for a total that includes a percent fee and a fixed…	**HSF-BF.A.1**

SET 65 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A ticket website charges a 2.5% fee on a $600 order.

What is the fee?

A. $25

B. $1.50

C. $15

D. $150

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 65 • MAIN • Q1/5

412

QUESTION ID

S65-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. $15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 65 • MAIN • Q1/5

A bank charges a 3% fee to transfer $1,250.

What is the fee?

Fee: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 65 • MAIN • Q2/5

413

QUESTION ID

S65-M-Q2

TYPE

Short answer

CORRECT ANSWER

$37.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 65 • MAIN • Q2/5

A concert ticket costs $85 plus a 12% service fee.

What is the total cost?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 65 • MAIN • Q3/5

414

QUESTION ID

S65-M-Q3

TYPE

Short answer

CORRECT ANSWER

$95.20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 65 • MAIN • Q3/5

True or false?

A 1.5% card fee on a $400 payment is $6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 65 • MAIN • Q4/5

415

QUESTION ID

S65-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 65 • MAIN • Q4/5

True or false?

A 4% fee on $250 is $1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 65 • MAIN • Q5/5

416

QUESTION ID

S65-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The fee is $10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 65 • MAIN • Q5/5

SET 65 • 7.RP.A.3

BACKWARD 1

Write a percent as a decimal • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write 1.5% as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 65 • BACKWARD 1 • Q1/5

418

QUESTION ID

S65-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.015

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 65 • BACKWARD 1 • Q1/5

Write 12% as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 65 • BACKWARD 1 • Q2/5

419

QUESTION ID

S65-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 65 • BACKWARD 1 • Q2/5

Which decimal is equal to 3%?

A. 3.0

B. 0.3

C. 0.003

D. 0.03

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 65 • BACKWARD 1 • Q3/5

420

QUESTION ID

S65-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0.03

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 65 • BACKWARD 1 • Q3/5

True or false?

0.5% = 0.005

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 65 • BACKWARD 1 • Q4/5

421

QUESTION ID

S65-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 65 • BACKWARD 1 • Q4/5

True or false?

0.4% = 0.4

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 65 • BACKWARD 1 • Q5/5

422

QUESTION ID

S65-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.4% = 0.004.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 65 • BACKWARD 1 • Q5/5

SET 65 • 7.RP.A.3

BACKWARD 2

Multiply decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

0.03 × 1,250

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 65 • BACKWARD 2 • Q1/5

424

QUESTION ID

S65-B2-Q1

TYPE

Short answer

CORRECT ANSWER

37.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 65 • BACKWARD 2 • Q1/5

Multiply.

1.12 × 85

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 65 • BACKWARD 2 • Q2/5

425

QUESTION ID

S65-B2-Q2

TYPE

Short answer

CORRECT ANSWER

95.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 65 • BACKWARD 2 • Q2/5

Multiply.

0.015×400

A. 6

B. 60

C. 4.15

D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 65 • BACKWARD 2 • Q3/5

QUESTION ID

S65-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 65 • BACKWARD 2 • Q3/5

True or false?

$0.025 \times 80 = 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 65 • BACKWARD 2 • Q4/5

427

QUESTION ID

S65-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 65 • BACKWARD 2 • Q4/5

True or false?

$0.12 \times 50 = 60$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 65 • BACKWARD 2 • Q5/5

428

QUESTION ID

S65-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 65 • BACKWARD 2 • Q5/5

SET 65 • 7.RP.A.3

FORWARD 1

Compare fee plans with equations that have the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Site A charges a $3 fee plus 1% of the amount a. Site B charges 2.5% of the amount a.

Solve 3 + 0.01a = 0.025a to find when the fees are equal.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 65 • FORWARD 1 • Q1/5

430

QUESTION ID

S65-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 65 • FORWARD 1 • Q1/5

Solve for a.

$0.04(a + 50) = 0.03a + 5$

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 65 • FORWARD 1 • Q2/5

431

QUESTION ID

S65-F1-Q2

TYPE

Short answer

CORRECT ANSWER

300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 65 • FORWARD 1 • Q2/5

Solve for x.

1.12x = x + 18

A. x = 18

B. x = 16.07

C. x = 1.5

D. x = 150

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 65 • FORWARD 1 • Q3/5

QUESTION ID

S65-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 150

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 65 • FORWARD 1 • Q3/5

True or false?

a = 500 is the solution of 2 + 0.02a = 0.024a.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 65 • FORWARD 1 • Q4/5

433

QUESTION ID

S65-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 65 • FORWARD 1 • Q4/5

True or false?

a = 200 is the solution of 5 + 0.03a = 0.08a.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 65 • FORWARD 1 • Q5/5

434

QUESTION ID

S65-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is a = 100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 65 • FORWARD 1 • Q5/5

SET 65 • 7.RP.A.3

FORWARD 2

Write a function for a total that includes a percent fee and a fixed charge • High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1

A ticket site adds a 10% fee and a $2.50 processing charge to a ticket price t.

Write a function T(t) for the total.

T(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 65 • FORWARD 2 • Q1/5

436

QUESTION ID

S65-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.1t + 2.5

GRADING NOTE t + 0.1t + 2.50 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q1

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 65 • FORWARD 2 • Q1/5

A ticket total is T(t) = 1.1t + 2.5.

What ticket price t gives a total of $68.50?

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 65 • FORWARD 2 • Q2/5

437

QUESTION ID

S65-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q2

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 65 • FORWARD 2 • Q2/5

A bank fee is 2% of the amount, but never less than $5.

What is the fee on $150?

A. $3

B. $5

C. $7

D. $2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 65 • FORWARD 2 • Q3/5

438

QUESTION ID

S65-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q3

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 65 • FORWARD 2 • Q3/5

True or false?

T(t) = 1.05t + 1 and T(t) = t + 0.05t + 1 are the same function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 65 • FORWARD 2 • Q4/5

439

QUESTION ID

S65-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q4

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 65 • FORWARD 2 • Q4/5

True or false?

For T(t) = 1.08t + 3, the fee part of the total is 1.08t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 65 • FORWARD 2 • Q5/5

440

QUESTION ID

S65-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The fee part is 0.08t + 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S65-F2-Q5

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1 • Set 65 • FORWARD 2 • Q5/5

SET 66 • 7.RP.A.3 • RATIOS & PROPORTIONAL RELATIONSHIPS

Solve multistep percent problems that combine a discount with tax or a tip

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Find a percent of an amount of money	**6.RP.A.3.c**
BACKWARD 2	Subtract amounts of money	**6.NS.B.3**
FORWARD 1	Solve multistep percent equations with the variable on both sides	**8.EE.C.7.b**
FORWARD 2	Construct an exponential function for repeated percent changes	**HSF-LE.A.2**

SET 66 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A bike costs $120. It is on sale for 25% off. Then 8% sales tax is added to the sale price.

What is the final cost?

Final cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 66 • MAIN • Q1/5

443

QUESTION ID

S66-M-Q1

TYPE

Short answer

CORRECT ANSWER

$97.20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 66 • MAIN • Q1/5

A meal costs $60. A $5 coupon is taken off the bill. Then a 20% tip on the $60 price is added.

What is the final amount paid?

Amount: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 66 • MAIN • Q2/5

444

QUESTION ID

S66-M-Q2

TYPE

Short answer

CORRECT ANSWER

$67

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 66 • MAIN • Q2/5

A $250 phone is 20% off. Then 6% sales tax is added to the sale price.

What is the final cost?

A. $200

B. $215

C. $212

D. $265

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 66 • MAIN • Q3/5

445

QUESTION ID

S66-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $212

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 66 • MAIN • Q3/5

True or false?

A $40 shirt is marked down 10%. Then the sale price is marked down another 10%. The final price is $32.40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 66 • MAIN • Q4/5

446

QUESTION ID

S66-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 66 • MAIN • Q4/5

True or false?

A 30% discount followed by 8% tax on the sale price is the same as a 22% discount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 66 • MAIN • Q5/5

447

QUESTION ID
S66-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 0.7 × 1.08 = 0.756, which is a 24.4% decrease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 66 • MAIN • Q5/5

SET 66 • 7.RP.A.3

BACKWARD 1

Find a percent of an amount of money • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 25% of $120?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 66 • BACKWARD 1 • Q1/5

449

QUESTION ID

S66-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 66 • BACKWARD 1 • Q1/5

What is 80% of $45?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 66 • BACKWARD 1 • Q2/5

450

QUESTION ID

S66-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 66 • BACKWARD 1 • Q2/5

What is 6% of $200?

A. $12

B. $120

C. $6

D. $1.20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 66 • BACKWARD 1 • Q3/5

451

QUESTION ID

S66-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 66 • BACKWARD 1 • Q3/5

True or false?

20% of $250 is $50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 66 • BACKWARD 1 • Q4/5

452

QUESTION ID

S66-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 66 • BACKWARD 1 • Q4/5

True or false?

30% of $90 is $30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 66 • BACKWARD 1 • Q5/5

453

QUESTION ID

S66-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 30% of $90 is $27.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 66 • BACKWARD 1 • Q5/5

SET 66 • 7.RP.A.3

BACKWARD 2

Subtract amounts of money • Grade 6 • The Number System • 6.NS.B.3

Subtract.

$84.50 − $12.68

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 66 • BACKWARD 2 • Q1/5

455

QUESTION ID

S66-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$71.82

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 66 • BACKWARD 2 • Q1/5

Subtract.

$65.00 − $9.75

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 66 • BACKWARD 2 • Q2/5

456

QUESTION ID

S66-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$55.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 66 • BACKWARD 2 • Q2/5

Subtract.

$100 − $37.45

A. $137.45

B. $62.45

C. $63.55

D. $62.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 66 • BACKWARD 2 • Q3/5

457

QUESTION ID

S66-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $62.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 66 • BACKWARD 2 • Q3/5

True or false?

$50.00 − $12.50 = $37.50

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 66 • BACKWARD 2 • Q4/5

458

QUESTION ID

S66-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 66 • BACKWARD 2 • Q4/5

True or false?

$75.20 − $8.40 = $67.20

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 66 • BACKWARD 2 • Q5/5

459

QUESTION ID

S66-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • 7.RP.A.3

FORWARD 1

Solve multistep percent equations with the variable on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Store A sells an item for 20% off its price p plus $10 shipping. Store B sells it for 10% off with free shipping.

Solve 0.8p + 10 = 0.9p to find when the stores charge the same.

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 66 • FORWARD 1 • Q1/5

461

QUESTION ID

S66-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 66 • FORWARD 1 • Q1/5

Solve for p.

$1.08(0.75p) = 0.75p + 4.5$

p = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 66 • FORWARD 1 • Q2/5

462

QUESTION ID

S66-F1-Q2

TYPE

Short answer

CORRECT ANSWER

75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 66 • FORWARD 1 • Q2/5

Solve for p.

1.06(0.9p) = 0.9p + 5.4

A. p = 90

B. p = 54

C. p = 100

D. p = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 66 • FORWARD 1 • Q3/5

463

QUESTION ID

S66-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. p = 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 66 • FORWARD 1 • Q3/5

True or false?

p = 200 is the solution of 0.7p + 15 = 0.75p + 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 66 • FORWARD 1 • Q4/5

464

QUESTION ID

S66-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 66 • FORWARD 1 • Q4/5

True or false?

p = 50 is the solution of 1.1(0.8p) = p − 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 66 • FORWARD 1 • Q5/5

465

QUESTION ID
S66-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 1.1(0.8)(50) = 44, but 50 − 8 = 42.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 66 • FORWARD 1 • Q5/5

SET 66 • 7.RP.A.3

FORWARD 2

Construct an exponential function for repeated percent changes • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

A $200 price is marked down 10% each week.

Write a function P(w) for the price after w weeks.

P(w) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 66 • FORWARD 2 • Q1/5

467

QUESTION ID

S66-F2-Q1

TYPE

Short answer

CORRECT ANSWER

200(0.9)^w

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 66 • FORWARD 2 • Q1/5

The price after w weeks is P(w) = 200(0.9)^w dollars.

Find the price after 2 weeks.

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 66 • FORWARD 2 • Q2/5

468

QUESTION ID

S66-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$162

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 66 • FORWARD 2 • Q2/5

A $500 item increases in price by 4% each year.

Which function gives its price after t years?

A. P(t) = 500 + 1.04t

B. P(t) = 500(1.04)^t

C. P(t) = 500 + 20t

D. P(t) = 500(0.04)^t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 66 • FORWARD 2 • Q3/5

469

QUESTION ID

S66-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. P(t) = 500(1.04)^t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 66 • FORWARD 2 • Q3/5

True or false?

A 20% discount followed by a 10% discount multiplies the price by 0.72.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 66 • FORWARD 2 • Q4/5

470

QUESTION ID

S66-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 66 • FORWARD 2 • Q4/5

True or false?

P(t) = 80(0.95)^t models a price that increases 95% each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 66 • FORWARD 2 • Q5/5

471

QUESTION ID
S66-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It decreases 5% each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S66-F2-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 66 • FORWARD 2 • Q5/5

SET 11 • 7.RP.A.3 • RATIOS & PROPORTIONAL RELATIONSHIPS

Find the percent error of an estimate or a measurement

MAIN	Grade 7 main question	**7.RP.A.3**
BACKWARD 1	Find the absolute value of a number	**6.NS.C.7.c**
BACKWARD 2	Write a fraction as a percent	**6.RP.A.3.c**
FORWARD 1	Compute with numbers in scientific notation — nearest related (n…	**8.EE.A.4**
FORWARD 2	Choose a level of accuracy that fits a measurement	**HSN-Q.A.3**

SET 11 • 7.RP.A.3

MAIN

Grade 7 main question • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

Ty estimates that 50 people are at a meeting. The actual number is 40.

What is the percent error?

Percent error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q1/5

474

QUESTION ID

S11-M-Q1

TYPE

Short answer

CORRECT ANSWER

25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q1/5

A scale shows 4.8 kg for a bag whose actual mass is 5 kg.

What is the percent error?

Percent error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q2/5

475

QUESTION ID

S11-M-Q2

TYPE

Short answer

CORRECT ANSWER

4%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q2/5

A student measures a line that is actually 20 cm long as 19 cm.

What is the percent error?

A. 95%

B. 5%

C. 5.3%

D. 1%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q3/5

476

QUESTION ID

S11-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q3/5

True or false?

A guess of 90 for an actual value of 100 has a percent error of 10%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q4/5

QUESTION ID

S11-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q4/5

True or false?

To find percent error, you divide the size of the error by the estimate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q5/5

478

QUESTION ID

S11-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE You divide by the actual value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-M-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • MAIN • Q5/5

SET 11 • 7.RP.A.3

BACKWARD 1

Find the absolute value of a number • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$|-12|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q1/5

480

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q1/5

Find the value.

$|-4.5|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q2/5

481

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q2/5

Which number has the greatest absolute value?

A. −3

B. 8

C. −9

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q3/5

482

QUESTION ID

S11-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. −9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q3/5

True or false?

$|-15| = 15$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q4/5

483

QUESTION ID

S11-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q4/5

True or false?

$|-2| < |1|$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q5/5

484

QUESTION ID

S11-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 11 • BACKWARD 1 • Q5/5

SET 11 • 7.RP.A.3

BACKWARD 2

Write a fraction as a percent • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

Write $\frac{3}{20}$ as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q1/5

486

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

15%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q1/5

Write $\frac{7}{25}$ as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q2/5

487

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

28%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q2/5

Which percent is equal to $\frac{9}{12}$?

A. 75%

B. 80%

C. 9%

D. 12%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q3/5

488

QUESTION ID

S11-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 75%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{8} = 12.5\%$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q4/5

489

QUESTION ID

S11-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q4/5

True or false?

$\frac{2}{5} = 25\%$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q5/5

490

QUESTION ID

S11-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • BACKWARD 2 • Q5/5

SET 11 • 7.RP.A.3

FORWARD 1

Compute with numbers in scientific notation • Grade 8 • Expressions & Equations • 8.EE.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Multiply. Write the answer in scientific notation.

$(3 \times 10^4) \times (2 \times 10^3)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q1/5 • NEAREST RELATED

492

QUESTION ID

S11-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6×10^7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q1/5 • NEAREST RELATED

Divide. Write the answer in scientific notation.

$(8 \times 10^{9}) \div (4 \times 10^{3})$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q2/5 • NEAREST RELATED

493

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2×10^{6}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q2/5 • NEAREST RELATED

Add.

$(2.5 \times 10^5) + (4 \times 10^4)$

A. 6.5×10^5

B. 2.9×10^5

C. 2.9×10^4

D. 6.5×10^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q3/5 • NEAREST RELATED

494

QUESTION ID

S11-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2.9×10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

$(5 \times 10^3) \times (3 \times 10^2) = 1.5 \times 10^6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q4/5 • NEAREST RELATED

495

QUESTION ID

S11-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

$(6 \times 10^8) \div (2 \times 10^2) = 3 \times 10^4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q5/5 • NEAREST RELATED

496

QUESTION ID

S11-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The quotient is 3×10^6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 11 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 11 • 7.RP.A.3

FORWARD 2

Choose a level of accuracy that fits a measurement • High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3

A ruler is marked in millimeters. A length is read as 12.4 cm.

To what place value is this measurement given?

Place: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q1/5

498

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Tenths of a centimeter (the nearest millimeter)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q1/5

A scale measures mass to the nearest 0.1 kg.

Which reading is reported to an accuracy that fits this scale?

A. 3 kg

B. 3.4 kg

C. 3.4172 kg

D. 3.41 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q2/5

QUESTION ID

S11-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 3.4 kg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The actual length of a track is 400 m. A measurement of it is off by at most 2 m.

What is the greatest possible percent error?

Percent error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q3/5

500

QUESTION ID

S11-F2-Q3

TYPE

Short answer

CORRECT ANSWER

0.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q3/5

True or false?

A length measured with a meter stick marked only in centimeters should not be reported as 1.2347 m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q4/5

501

QUESTION ID

S11-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q4/5

True or false?

Writing more decimal places always makes a measurement more accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q5/5

502

QUESTION ID

S11-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S11-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: Quantities • HSN-Q.A.3 • Set 11 • FORWARD 2 • Q5/5

SET 12 • 7.NS.A.1.a • THE NUMBER SYSTEM

Describe situations in which opposite quantities combine to make 0

MAIN	Grade 7 main question	**7.NS.A.1.a**
BACKWARD 1	Use positive and negative numbers to represent quantities	**6.NS.C.5**
BACKWARD 2	Recognize opposite numbers	**6.NS.C.6.a**
FORWARD 1	Use opposite exponents with the properties of integer exponents…	**8.EE.A.1**
FORWARD 2	Justify each step in solving an equation and find steps that chang…	**HSA-REI.A.1**

SET 12 • 7.NS.A.1.a

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.a

Ari has $15 in his wallet. He spends all $15.

Write an addition expression for this situation and find its value.

Expression and value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q1/5

505

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

15 + (−15) = 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q1/5

Which situation combines to make 0?

A. Gaining 5 points, then gaining 5 more points

B. A temperature drop of 6°, then another drop of 6°

C. Climbing up 8 floors, then down 8 floors

D. Walking 3 miles east, then 2 miles west

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q2/5

506

QUESTION ID

S12-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Climbing up 8 floors, then down 8 floors

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q2/5

A diver is 12 feet below the surface of the water (−12 feet).

How many feet must she rise to be at the surface (0 feet)?

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q3/5

507

QUESTION ID

S12-M-Q3

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-M-Q3

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q3/5

True or false?

A gain of 9 yards followed by a loss of 9 yards is a total change of 0 yards.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q4/5

508

QUESTION ID

S12-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q4/5

True or false?

A deposit of $40 followed by a withdrawal of $30 leaves an account balance unchanged.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q5/5

509

QUESTION ID

S12-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 7 • The Number System • 7.NS.A.1.a • Set 12 • MAIN • Q5/5

SET 12 • 7.NS.A.1.a

BACKWARD 1

Use positive and negative numbers to represent quantities • Grade 6 • The Number System • 6.NS.C.5

Write an integer for 20 feet below sea level.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q1/5

511

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q1/5

Write an integer for a gain of 7 yards.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q2/5

512

QUESTION ID

S12-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE +7 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q2/5

Which integer represents a withdrawal of $35?

A. 35

B. −35

C. −3.5

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q3/5

513

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q3/5

True or false?

A temperature of 4 degrees below zero can be written as −4°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q4/5

514

QUESTION ID

S12-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q4/5

True or false?

The integer −10 represents a deposit of $10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q5/5

515

QUESTION ID

S12-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • 7.NS.A.1.a

BACKWARD 2

Recognize opposite numbers • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 8?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q1/5

517

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q1/5

What is the opposite of −13?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q2/5

518

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

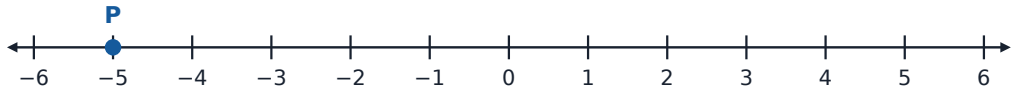
13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q2/5

Point P is at −5.

Which number is the same distance from 0 on the other side of 0?



A. 5 **B.** −10 **C.** 10 **D.** 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q3/5

519

QUESTION ID

S12-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q3/5

True or false?

The opposite of the opposite of 6 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q4/5

520

QUESTION ID

S12-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q4/5

True or false?

0 and 1 are opposites.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q5/5

521

QUESTION ID

S12-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 12 • BACKWARD 2 • Q5/5

SET 12 • 7.NS.A.1.a

FORWARD 1

Use opposite exponents with the properties of integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Find the value.

$5^{3} \times 5^{-3}$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q1/5 • NEAREST RELATED

523

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q1/5 • NEAREST RELATED

Write as a single power. Then find its value.

$2^4 \times 2^{-6}$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q2/5 • NEAREST RELATED

524

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$2^{-2} = 1/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q2/5 • NEAREST RELATED

Which expression is equal to 1?

A. $7^2 \times 7^2$

B. $7^{-2} \times 7^{-2}$

C. $7^{-2} + 7^2$

D. $7^{-2} \times 7^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q3/5 • NEAREST RELATED

525

QUESTION ID

S12-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $7^{-2} \times 7^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

$10^{-2} = \frac{1}{100}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q4/5 • NEAREST RELATED

526

QUESTION ID

S12-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

$3^{-2} = -9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q5/5 • NEAREST RELATED

527

QUESTION ID

S12-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^{-2} = 1/9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 12 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 12 • 7.NS.A.1.a

FORWARD 2

Justify each step in solving an equation and find steps that change the solution • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

Solve $\frac{1}{2}(x - 6) = 4 + x$. Write each step and the reason for it.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q1/5

529

QUESTION ID

S12-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

−14

GRADING NOTE

REQUIRED WORK: Every step with a valid reason. Example: distributive property ($\frac{1}{2}x - 3 = 4 + x$); subtract x from both sides ($-\frac{1}{2}x - 3 = 4$); add 3 to both sides ($-\frac{1}{2}x = 7$); multiply both sides by −2 ($x = -14$). Other valid orders are accepted. A correct answer without reasons is not full credit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S12-F2-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q1/5

Lia solved $3(x - 4) = 2x + 5$ like this:

Step 1: $3x - 4 = 2x + 5$

Step 2: $x = 9$

Which step is wrong? Give the correct solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q2/5

530

QUESTION ID

S12-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Step 1 is wrong: $3(x - 4) = 3x - 12$. The correct solution is $x = 17$.

GRADING NOTE Grade both: Step 1 identified with the reason, and $x = 17$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q2/5

Which step can change the solutions of an equation?

A. Dividing both sides of $-4x = 12$ by -4

B. Multiplying both sides of $\frac{x}{3} = 5$ by 3

C. Multiplying both sides of $2x = 6$ by 0

D. Adding -7 to both sides of $x + 7 = 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q3/5

531

QUESTION ID

S12-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Multiplying both sides of 2x = 6 by 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q3/5

True or false?

Adding 2x to both sides of 5 − 2x = x + 11 gives 5 = 3x + 11, which has the same solution as the original equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q4/5

532

QUESTION ID

S12-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q4/5

True or false?

Squaring both sides of $x = 3$ gives $x^2 = 9$, which has the same solutions as $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q5/5

533

QUESTION ID

S12-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 = 9$ also has the solution $x = -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S12-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 12 • FORWARD 2 • Q5/5

SET 13 • 7.NS.A.1.b • THE NUMBER SYSTEM

Add rational numbers on a number line

MAIN	Grade 7 main question	**7.NS.A.1.b**
BACKWARD 1	Find and position integers on a number line	**6.NS.C.6.c**
BACKWARD 2	Add decimals	**5.NBT.B.7**
FORWARD 1	Locate irrational numbers on a number line	**8.NS.A.2**
FORWARD 2	Decide whether a sum is rational or irrational	**HSN-RN.B.3**

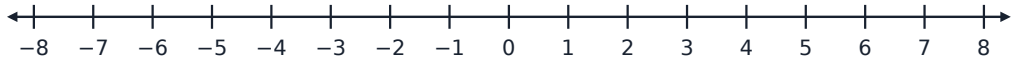
SET 13 • 7.NS.A.1.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.b

Use the number line to add.

$-3 + 7$



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q1/5

536

QUESTION ID

S13-M-Q1

TYPE

Short answer

CORRECT ANSWER

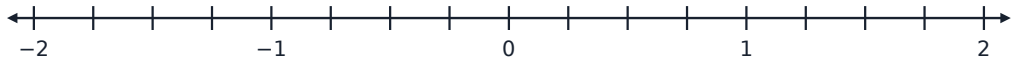
4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q1/5

Use the number line to add.

$-\frac{3}{4} + 1\frac{1}{2}$



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q2/5

537

QUESTION ID

S13-M-Q2

TYPE

Short answer

CORRECT ANSWER

3/4

GRADING NOTE 0.75 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q2/5

Start at 2 on a number line and move 5 units to the left.

Which sum does this show?

A. −5 + (−2)

B. −2 + 5

C. 2 + 5

D. 2 + (−5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q3/5

538

QUESTION ID

S13-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2 + (−5)

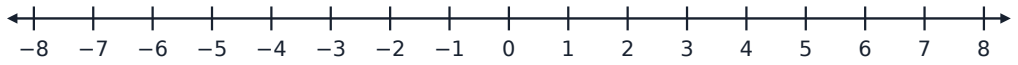
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q3/5

Show −2 + 6 on the number line. Start at −2 and draw an arrow 6 units long.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q4/5

539

QUESTION ID

S13-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: An arrow that starts at −2 and ends at 4 (6 units to the right)

Written: 4

GRADING NOTE Grade both: the arrow from −2 to 4, and the sum 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q4/5

True or false?

$$-\frac{3}{4} + \frac{5}{6} = \frac{1}{12}$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q5/5

540

QUESTION ID

S13-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

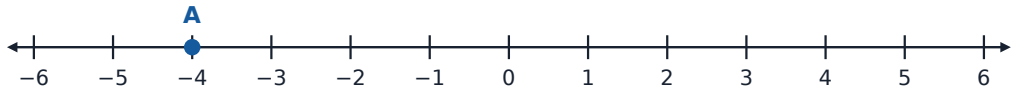
Grade 7 • The Number System • 7.NS.A.1.b • Set 13 • MAIN • Q5/5

SET 13 • 7.NS.A.1.b

BACKWARD 1

Find and position integers on a number line • Grade 6 • The Number System • 6.NS.C.6.c

What number is at point A?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q1/5

542

QUESTION ID

S13-B1-Q1

TYPE

Short answer

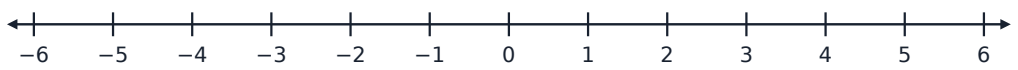
CORRECT ANSWER

−4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q1/5

Plot −3 and 5 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q2/5

543

QUESTION ID

S13-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Points at −3 and 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q2/5

Which number is farthest to the left on a number line?

A. −2

B. 0

C. 3

D. −7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q3/5

544

QUESTION ID

S13-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q3/5

True or false?

−1 is to the right of −5 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q4/5

545

QUESTION ID

S13-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q4/5

True or false?

−6 is between −3 and 0 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q5/5

546

QUESTION ID

S13-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 13 • BACKWARD 1 • Q5/5

SET 13 • 7.NS.A.1.b

BACKWARD 2

Add decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Add.

3.75 + 2.5

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q1/5

548

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q1/5

Add.

0.8 + 4.65

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q2/5

549

QUESTION ID

S13-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q2/5

Add.

12.4 + 3.06

A. 46.4

B. 15.10

C. 12.71

D. 15.46

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q3/5

550

QUESTION ID

S13-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 15.46

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q3/5

True or false?

$2.7 + 1.3 = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q4/5

551

QUESTION ID

S13-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q4/5

True or false?

5.6 + 0.04 = 6

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q5/5

552

QUESTION ID

S13-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • 7.NS.A.1.b

FORWARD 1

Locate irrational numbers on a number line • Grade 8 • The Number System • 8.NS.A.2

Between which two whole numbers is √20?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q1/5

554

QUESTION ID

S13-F1-Q1

TYPE

Short answer

CORRECT ANSWER

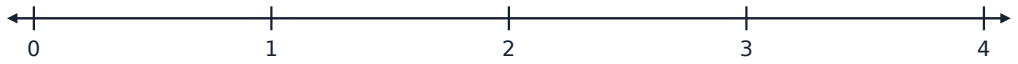
4 and 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q1/5

Plot an approximate location of √2 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q2/5

555

QUESTION ID

S13-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A point between 1 and 2, a little less than halfway (about 1.4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q2/5

Which is the best approximation of √50 to the nearest tenth?

A. 25

B. 7.1

C. 7.5

D. 7.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q3/5

556

QUESTION ID

S13-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F1-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q3/5

True or false?

√10 is between 3 and 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q4/5

557

QUESTION ID

S13-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q4/5

True or false?

π is less than 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q5/5

558

QUESTION ID

S13-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 13 • FORWARD 1 • Q5/5

SET 13 • 7.NS.A.1.b

FORWARD 2

Decide whether a sum is rational or irrational • High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3

Is 3 + √2 rational or irrational?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q1/5

560

QUESTION ID

S13-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Irrational

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q1

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q1/5

Is $\frac{1}{2} + \frac{3}{4}$ rational or irrational? Give its value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q2/5

561

QUESTION ID

S13-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Rational; 5/4

GRADING NOTE 1 1/4 or 1.25 is also correct for the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q2

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q2/5

Which sum is irrational?

A. $\sqrt{9} + 4$

B. $0.5 + \frac{1}{2}$

C. $-2 + 2$

D. $\sqrt{3} + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q3/5

562

QUESTION ID

S13-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $\sqrt{3} + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q3

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q3/5

True or false?

The sum of two rational numbers is always rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q4/5

563

QUESTION ID

S13-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q4

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q4/5

True or false?

The sum of a rational number and an irrational number can be rational.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q5/5

564

QUESTION ID

S13-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S13-F2-Q5

High School (Grade 9 in this bank) • Number & Quantity: The Real Number System • HSN-RN.B.3 • Set 13 • FORWARD 2 • Q5/5

SET 14 • 7.NS.A.1.b • THE NUMBER SYSTEM

Use additive inverses: a number plus its opposite is 0

MAIN	Grade 7 main question	**7.NS.A.1.b**
BACKWARD 1	Opposites are the same distance from 0	**6.NS.C.6.a**
BACKWARD 2	Interpret absolute value as distance from 0	**6.NS.C.7.c**
FORWARD 1	Tell whether a linear equation has one, no, or infinitely many solut…	**8.EE.C.7.a**
FORWARD 2	Solve linear equations with letter coefficients or fraction coefficien…	**HSA-REI.B.3**

SET 14 • 7.NS.A.1.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.b

What number added to −9.5 gives 0?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q1/5

567

QUESTION ID

S14-M-Q1

TYPE

Short answer

CORRECT ANSWER

9.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q1/5

Fill in the blank.

$\frac{3}{4}$ + ___ = 0

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q2/5

568

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

−3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q2/5

Which sum is equal to 0?

A. $-2.6 + (-2.6)$

B. $-2.6 + 2.6$

C. $2.6 + 0$

D. $-2.6 + 6.2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q3/5

569

QUESTION ID

S14-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −2.6 + 2.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q3/5

True or false?

For any rational number p, −p + p = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q4/5

570

QUESTION ID

S14-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q4/5

True or false?

The additive inverse of -7 is -7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q5/5

571

QUESTION ID

S14-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The additive inverse of -7 is 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 14 • MAIN • Q5/5

SET 14 • 7.NS.A.1.b

BACKWARD 1

Opposites are the same distance from 0 • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 2.5?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q1/5

573

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q1/5

Which pair of numbers are opposites?

A. −11 and 11

B. −11 and −11

C. 11 and $\frac{1}{11}$

D. 1 and −11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q2/5

574

QUESTION ID

S14-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. −11 and 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q2/5

How far is −9 from 0 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q3/5

575

QUESTION ID

S14-B1-Q3

TYPE

Short answer

CORRECT ANSWER

9 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q3/5

True or false?

$-\frac{2}{3}$ and $\frac{2}{3}$ are the same distance from 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q4/5

576

QUESTION ID

S14-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q4/5

True or false?

The opposite of 0 is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q5/5

577

QUESTION ID

S14-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 14 • BACKWARD 1 • Q5/5

SET 14 • 7.NS.A.1.b

BACKWARD 2

Interpret absolute value as distance from 0 • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$|-18|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q1/5

579

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q1/5

A diver is at −30 feet.

How far is she from the surface (0 feet)?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q2/5

580

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

30 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q2/5

Which number has an absolute value of 6?

A. $\frac{1}{6}$

B. -6

C. $-\frac{1}{6}$

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q3/5

581

QUESTION ID

S14-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q3/5

True or false?

$|-3.5| = |3.5|$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q4/5

582

QUESTION ID

S14-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q4/5

True or false?

$|-20| = -20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q5/5

583

QUESTION ID

S14-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 14 • BACKWARD 2 • Q5/5

SET 14 • 7.NS.A.1.b

FORWARD 1

Tell whether a linear equation has one, no, or infinitely many solutions • Grade 8 • Expressions & Equations • 8.EE.C.7.a

How many solutions does this equation have?

$3x + 5 = 3x - 2$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q1/5

585

QUESTION ID

S14-F1-Q1

TYPE

Short answer

CORRECT ANSWER

No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q1/5

How many solutions does this equation have?

$2(x + 4) = 2x + 8$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q2/5

586

QUESTION ID

S14-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q2/5

Which equation has exactly one solution?

A. $4x + 1 = 2x + 7$

B. $2(2x) = 4x$

C. $4x + 1 = 4x + 7$

D. $4x + 1 = 4x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q3/5

587

QUESTION ID

S14-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $4x + 1 = 2x + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q3/5

True or false?

$x + 6 = x - 6$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q4/5

588

QUESTION ID

S14-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q4/5

True or false?

$5x - 5x = 0$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q5/5

589

QUESTION ID

S14-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is true for every x, so it has infinitely many solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 14 • FORWARD 1 • Q5/5

SET 14 • 7.NS.A.1.b

FORWARD 2

Solve linear equations with letter coefficients or fraction coefficients • High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve for x. Assume a ≠ 0.

ax + 5 = 2

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q1

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q1/5

591

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−3/a

GRADING NOTE $\frac{-3}{a}$ written in any equivalent form is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F2-Q1**

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q1/5

Solve for x. Assume a ≠ b.

ax + c = bx

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q2/5

592

QUESTION ID

S14-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−c/(a − b)

GRADING NOTE c/(b − a) is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q2

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q2/5

Solve for x.

$\frac{2}{3}(x - 4) = \frac{1}{4}x + 1$

A. $x = \frac{44}{5}$

B. $x = 12$

C. $x = \frac{20}{3}$

D. $x = \frac{11}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q3/5

593

QUESTION ID

S14-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 44/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q3

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q3/5

True or false?

If $k \neq 0$, the solution of $k(x - 2) = 3k$ is $x = 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q4/5

QUESTION ID

S14-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q4

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q4/5

True or false?

The solution of $mx + 4 = 4$ is $x = 1$ for every nonzero value of m.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q5/5

595

QUESTION ID

S14-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S14-F2-Q5

High School (Grade 9 in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • FORWARD 2 • Q5/5

SET 15 • 7.NS.A.1.b • THE NUMBER SYSTEM

Interpret sums of rational numbers in real-world situations

MAIN	Grade 7 main question	**7.NS.A.1.b**
BACKWARD 1	Use signed numbers in a real-world situation	**6.NS.C.5**
BACKWARD 2	Compare signed numbers in a real-world situation	**6.NS.C.7.b**
FORWARD 1	Model a changing quantity with a linear function	**8.F.B.4**
FORWARD 2	Evaluate a function and interpret the value in context	**HSF-IF.A.2**

SET 15 • 7.NS.A.1.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.b

The temperature was −5°F at dawn. It rose 12°F by noon.

What was the temperature at noon?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q1/5

598

QUESTION ID

S15-M-Q1

TYPE

Short answer

CORRECT ANSWER

7°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A stock's price changed by $-1\frac{1}{4}$ dollars on Monday and by $+2\frac{3}{4}$ dollars on Tuesday.

What was the total change over the two days?

Total change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q2/5

599

QUESTION ID
S15-M-Q2

TYPE
Short answer

CORRECT ANSWER
+1 1/2 dollars

GRADING NOTE +$1.50 or 3/2 dollars is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q2/5

Jo's account balance is −$18. She deposits $25.

Which sum gives her new balance?

A. −18 + (−25) = −43

B. 18 + 25 = 43

C. −18 + 25 = 7

D. 18 + (−25) = −7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q3/5

600

QUESTION ID

S15-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. −18 + 25 = 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q3/5

True or false?

A pond's water level changed by $-\frac{2}{3}$ foot one day and by $+\frac{1}{2}$ foot the next day. The total change was $-\frac{1}{6}$ foot.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q4/5

601

QUESTION ID

S15-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q4/5

True or false?

A hiker 350 feet below sea level climbs 400 feet. She ends 50 feet below sea level.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q5/5

602

QUESTION ID

S15-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE She ends 50 feet above sea level.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-M-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 15 • MAIN • Q5/5

SET 15 • 7.NS.A.1.b

BACKWARD 1

Use signed numbers in a real-world situation • Grade 6 • The Number System • 6.NS.C.5

A stock's price changed by −$3.25.

Was this a gain or a loss, and of how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q1/5

604

QUESTION ID

S15-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A loss of $3.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q1/5

Which number describes 15 meters above sea level?

A. $\frac{1}{15}$

B. 15

C. 0

D. -15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q2/5

605

QUESTION ID

S15-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q2/5

Write a number for a temperature of 8 degrees below zero.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q3/5

606

QUESTION ID

S15-B1-Q3

TYPE

Short answer

CORRECT ANSWER

−8°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q3/5

True or false?

In a bank account, a balance of −$50 can mean $50 is owed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q4/5

607

QUESTION ID

S15-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q4/5

True or false?

A temperature of −2° is warmer than 0°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q5/5

608

QUESTION ID

S15-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • 7.NS.A.1.b

BACKWARD 2

Compare signed numbers in a real-world situation • Grade 6 • The Number System • 6.NS.C.7.b

Which temperature is colder, −8°C or −3°C?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q1/5

610

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−8°C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q1/5

Which statement is true?

A. $-7 < -12$

B. $-12 > -7$

C. $-12 < -7$

D. $-7 = -12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q2/5

611

QUESTION ID

S15-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $-12 < -7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an inequality to compare elevations of −15 feet and −9 feet.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q3/5

612

QUESTION ID

S15-B2-Q3

TYPE

Short answer

CORRECT ANSWER

−15 < −9

GRADING NOTE −9 > −15 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q3/5

True or false?

A balance of −$20 is less than a balance of −$5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q4/5

613

QUESTION ID

S15-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q4/5

True or false?

−4°F is colder than −10°F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q5/5

614

QUESTION ID

S15-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 15 • BACKWARD 2 • Q5/5

SET 15 • 7.NS.A.1.b

FORWARD 1

Model a changing quantity with a linear function • Grade 8 • Functions • 8.F.B.4

A diver starts at −10 feet and descends 4 feet per minute.

Write a function for her position d after t minutes.

d = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q1/5

616

QUESTION ID

S15-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−4t − 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q1/5

The temperature is −6°F and rises 3°F per hour.

Write a function for the temperature T after h hours.

T = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q2/5

617

QUESTION ID

S15-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3h − 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q2/5

An account balance starts at −$40 and goes up $15 per week.

What is the balance after 4 weeks?

A. $20

B. −$100

C. −$20

D. $60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q3/5

618

QUESTION ID

S15-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q3/5

True or false?

For T = 3h − 6, the temperature reaches 0°F after 2 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q4/5

619

QUESTION ID

S15-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q4/5

True or false?

For d = −4t − 10, the starting position is −4 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q5/5

620

QUESTION ID

S15-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The starting position is −10 feet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 15 • FORWARD 1 • Q5/5

SET 15 • 7.NS.A.1.b

FORWARD 2

Evaluate a function and interpret the value in context • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2

$h(t) = -16t^2 + 64t$ gives the height, in feet, of a ball after t seconds.

Find h(1).

h(1) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q1/5

622

QUESTION ID

S15-F2-Q1

TYPE

Short answer

CORRECT ANSWER

48 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q1/5

T(h) = −2 + 1.5h gives the temperature in °C after h hours.

Find T(6). What does it mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q2/5

623

QUESTION ID

S15-F2-Q2

TYPE

Short answer

CORRECT ANSWER

7; the temperature is 7°C after 6 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q2/5

$f(x) = 3x - 8$

What is $f(-2)$?

A. 14

B. -14

C. -2

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q3/5

624

QUESTION ID

S15-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q3/5

True or false?

If $g(x) = -x + 5$, then $g(-3) = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q4/5

625

QUESTION ID

S15-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q4/5

True or false?

If $f(x) = x^2 - 4$, then $f(-3) = -13$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q5/5

626

QUESTION ID

S15-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f(-3) = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S15-F2-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • FORWARD 2 • Q5/5

SET 16 • 7.NS.A.1.c • THE NUMBER SYSTEM

Subtract rational numbers by adding the additive inverse

MAIN	Grade 7 main question	**7.NS.A.1.c**
BACKWARD 1	Subtract decimals	**5.NBT.B.7**
BACKWARD 2	Subtract fractions with unlike denominators	**5.NF.A.1**
FORWARD 1	Find a rate of change from two points with negative coordinates	**8.F.B.4**
FORWARD 2	Subtract polynomials	**HSA-APR.A.1**

SET 16 • 7.NS.A.1.c

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.c

Rewrite as an addition expression. Then find the value.

$4 - 9$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q1/5

629

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

$4 + (-9) = -5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q1/5

Subtract.

$-7 - (-12)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q2/5

630

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q2/5

Which expression is equal to $-3 - 8$?

A. $3 + (-8)$

B. $3 + 8$

C. $-3 + 8$

D. $-3 + (-8)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q3/5

631

QUESTION ID

S16-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −3 + (−8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q3/5

True or false?

$-2.5 - 4 = -6.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q4/5

632

QUESTION ID

S16-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q4/5

True or false?

$\frac{1}{2} - (-\frac{1}{4}) = \frac{1}{4}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q5/5

633

QUESTION ID

S16-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The difference is 3/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • MAIN • Q5/5

SET 16 • 7.NS.A.1.c

BACKWARD 1

Subtract decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Subtract.

8.2 − 3.75

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q1/5

635

QUESTION ID

S16-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q1/5

Subtract.

10 − 6.38

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q2/5

636

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3.62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q2/5

Subtract.

$5.4 - 2.9$

A. 8.3

B. 2.5

C. 3.5

D. 2.15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q3/5

637

QUESTION ID

S16-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q3/5

True or false?

$7.06 - 2.5 = 4.56$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q4/5

638

QUESTION ID

S16-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q4/5

True or false?

$9.3 - 0.7 = 9.6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q5/5

639

QUESTION ID

S16-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • 7.NS.A.1.c

BACKWARD 2

Subtract fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$\frac{5}{6} - \frac{1}{4}$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q1/5

641

QUESTION ID

S16-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q1/5

Subtract.

$3\frac{1}{2} - 1\frac{2}{3}$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q2/5

642

QUESTION ID

S16-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1 5/6

GRADING NOTE 11/6 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q2/5

Subtract.

$\frac{3}{4} - \frac{2}{5}$

A. $\frac{7}{20}$

B. $\frac{1}{20}$

C. $\frac{1}{2}$

D. $\frac{5}{9}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q3/5

643

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 7/20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q3/5

True or false?

$\frac{2}{3} - \frac{1}{6} = \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q4/5

644

QUESTION ID

S16-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q4/5

True or false?

$\frac{4}{5} - \frac{1}{3} = \frac{3}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q5/5

645

QUESTION ID

S16-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • 7.NS.A.1.c

FORWARD 1

Find a rate of change from two points with negative coordinates • Grade 8 • Functions • 8.F.B.4

Find the rate of change of the line through (−2, 5) and (3, −5).

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q1/5

647

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q1/5

Find the slope of the line through (−4, −1) and (2, 8).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q2/5

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3/2

GRADING NOTE 1.5 or 1 1/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q2/5

What is the slope of the line through (1, −3) and (−1, 5)?

A. 1

B. 4

C. −1

D. −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q3/5

649

QUESTION ID

S16-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q3/5

True or false?

The slope of the line through $(-3, -2)$ and $(0, 4)$ is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q4/5

650

QUESTION ID

S16-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q4/5

True or false?

The slope of the line through (2, −1) and (5, −7) is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q5/5

651

QUESTION ID

S16-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The slope is −2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 16 • FORWARD 1 • Q5/5

SET 16 • 7.NS.A.1.c

FORWARD 2

Subtract polynomials • High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Subtract.

$(5x^2 - 3x + 2) - (2x^2 - 7x - 4)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q1

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q1/5

653

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$3x^2 + 4x + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q1**

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q1/5

Subtract.

$(4x - 9) - (-x + 3)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q2

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q2/5

654

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5x − 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q2**

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q2/5

Subtract.

$(x^2 + 2x) - (3x^2 - 5x)$

A. $-2x^2 + 7x$

B. $-2x^2 - 3x$

C. $-2x^2 + 3x$

D. $4x^2 + 7x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q3

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q3/5

655

QUESTION ID

S16-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $-2x^2 + 7x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q3**

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q3/5

True or false?

$(6x + 1) - (2x - 5) = 4x + 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q4/5

656

QUESTION ID

S16-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q4

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q4/5

True or false?

$(3x^2 - x) - (x^2 + x) = 2x^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q5/5

657

QUESTION ID

S16-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The difference is $2x^2 - 2x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S16-F2-Q5

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • FORWARD 2 • Q5/5

SET 17 • 7.NS.A.1.c • THE NUMBER SYSTEM

Find the distance between two rational numbers as the absolute value of their difference

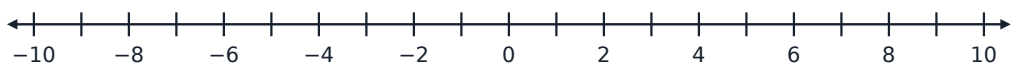
MAIN	Grade 7 main question	**7.NS.A.1.c**
BACKWARD 1	Find the distance between points with the same x- or y-coordinate	**6.NS.C.8**
BACKWARD 2	Position fractions and decimals on a number line	**6.NS.C.6.c**
FORWARD 1	Use the Pythagorean theorem to find the distance between two po…	**8.G.B.8**
FORWARD 2	Use coordinates to find the perimeter of a polygon	**HSG-GPE.B.7**

SET 17 • 7.NS.A.1.c

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.c

Find the distance between −6 and 9 on the number line.



Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q1/5

660

QUESTION ID

S17-M-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q1/5

Find the distance between −4.5 and −1.2 on a number line.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q2/5

661

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

3.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q2/5

Which expression gives the distance between −3 and 8 on a number line?

A. −3 − 8

B. |−3 − 8|

C. −3 + 8

D. |−3 + 8|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q3/5

662

QUESTION ID

S17-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. |−3 − 8|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q3/5

True or false?

$|2 - (-5)| = 7$, so 2 and -5 are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q4/5

663

QUESTION ID

S17-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q4/5

True or false?

The distance between −10 and −2 on a number line is 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q5/5

664

QUESTION ID

S17-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The distance is 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-M-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • MAIN • Q5/5

SET 17 • 7.NS.A.1.c

BACKWARD 1

Find the distance between points with the same x- or y-coordinate • Grade 6 • The Number System • 6.NS.C.8

Find the distance between (−3, 4) and (5, 4).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q1/5

666

QUESTION ID

S17-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q1/5

Find the distance between (2, −6) and (2, 1).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q2/5

667

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q2/5

Which two points are 5 units apart?

A. (1, −2) and (1, 3)

B. (1, −2) and (6, 3)

C. (−2, 1) and (2, 1)

D. (1, −2) and (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q3/5

668

QUESTION ID

S17-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (1, −2) and (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q3/5

True or false?

(−4, −1) and (−4, 6) are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q4/5

669

QUESTION ID

S17-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q4/5

True or false?

(−5, 2) and (3, 2) are 2 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q5/5

670

QUESTION ID

S17-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are 8 units apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B1-Q5

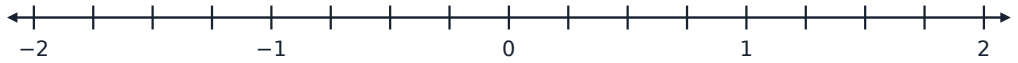
Grade 6 • The Number System • 6.NS.C.8 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • 7.NS.A.1.c

BACKWARD 2

Position fractions and decimals on a number line • Grade 6 • The Number System • 6.NS.C.6.c

Plot -1.5 and $\frac{3}{4}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q1/5

672

QUESTION ID

S17-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at −1.5 and 3/4 (0.75)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q1/5

Name a number between −2 and −1.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q2/5

673

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−1.5

GRADING NOTE Any number greater than −2 and less than −1 is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q2/5

Which number is between −0.5 and 0?

A. −0.75

B. −1

C. 0.25

D. −0.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q3/5

674

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −0.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q3/5

True or false?

$-\frac{1}{2}$ is to the left of 0 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q4/5

675

QUESTION ID

S17-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q4/5

True or false?

−2.3 is to the right of −2 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q5/5

676

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 17 • BACKWARD 2 • Q5/5

SET 17 • 7.NS.A.1.c

FORWARD 1

Use the Pythagorean theorem to find the distance between two points • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q1/5

678

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q1/5

Find the distance between (−2, −3) and (4, 5).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q2/5

679

QUESTION ID

S17-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q2/5

What is the distance between (0, 0) and (5, 12)?

A. 13 units

B. 7 units

C. 60 units

D. 17 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q3/5

680

QUESTION ID

S17-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 13 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q3/5

True or false?

The distance between (−1, 1) and (2, 5) is 5 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q4/5

681

QUESTION ID

S17-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q4/5

True or false?

The distance between (0, 0) and (3, 3) is 6 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q5/5

682

QUESTION ID

S17-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The distance is √18, about 4.24 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 17 • FORWARD 1 • Q5/5

SET 17 • 7.NS.A.1.c

FORWARD 2

Use coordinates to find the perimeter of a polygon • High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

A triangle has vertices (0, 0), (6, 0), and (6, 8).

Find its perimeter.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q1

High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q1/5

684

QUESTION ID

S17-F2-Q1

TYPE

Short answer

CORRECT ANSWER

24 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q1**

High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q1/5

A rectangle has vertices (−3, −2), (5, −2), (5, 4), and (−3, 4).

Find its perimeter.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q2

High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q2/5

685

QUESTION ID

S17-F2-Q2

TYPE

Short answer

CORRECT ANSWER

28 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q2**

High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q2/5

A triangle has vertices (−2, 1), (2, 1), and (2, 4).

What is its perimeter?

A. 10 units

B. 6 units

C. 12 units

D. 7 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q3

High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q3/5

686

QUESTION ID

S17-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 12 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S17-F2-Q3**

High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q3/5

True or false?

A square with vertices (1, 1), (4, 1), (4, 4), and (1, 4) has a perimeter of 12 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q4/5

687

QUESTION ID

S17-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q4

High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q4/5

True or false?

A triangle with vertices (0, 0), (8, 0), and (0, 6) has a perimeter of 14 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q5/5

688

QUESTION ID

S17-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The perimeter is 24 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S17-F2-Q5

High School (Grade 9 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 17 • FORWARD 2 • Q5/5

SET 18 • 7.NS.A.1.d • THE NUMBER SYSTEM

Use properties of operations to add and subtract rational numbers

MAIN	Grade 7 main question	**7.NS.A.1.d**
BACKWARD 1	Apply properties to write equivalent expressions	**6.EE.A.3**
BACKWARD 2	Add multi-digit decimals	**6.NS.B.3**
BACKWARD 3	Reorder and regroup addends to add within 100	**2.NBT.B.5**
FORWARD 1	Solve equations by collecting like terms	**8.EE.C.7.b**
FORWARD 2	Use the structure of an expression to rewrite it	**HSA-SSE.A.2**

SET 18 • 7.NS.A.1.d

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.1.d

Find the sum by grouping numbers to make the work easier. Show your work.

$-8 + 13.5 + 8$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q1/5

691

QUESTION ID

S18-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

13.5

GRADING NOTE REQUIRED WORK: Any grouping that uses the commutative and associative properties correctly. Example: $(-8 + 8) + 13.5 = 0 + 13.5$. Other valid groupings are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q1/5

Rewrite each subtraction as adding the opposite. Then use properties to find the value. Show your work.

$5.6 - 8 - 3.6 + 8$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q2/5

692

QUESTION ID

S18-M-Q2

TYPE

Answer with required work

CORRECT ANSWER

2

GRADING NOTE REQUIRED WORK: Subtraction rewritten as adding the opposite, then any valid regrouping. Example: $5.6 + (-8) + (-3.6) + 8 = (5.6 + (-3.6)) + (-8 + 8) = 2 + 0$. Other valid groupings are accepted; subtraction itself is not reordered.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q2/5

Which property is shown?

$-5 + 2.7 = 2.7 + (-5)$

A. Associative property of addition

B. Additive inverse property

C. Commutative property of addition

D. Additive identity property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q3/5

693

QUESTION ID

S18-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Commutative property of addition

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q3/5

True or false?

$-6 + (4 + 6) = (-6 + 6) + 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q4/5

694

QUESTION ID

S18-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$7 - 3 = 3 - 7$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q5/5

695

QUESTION ID

S18-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Subtraction is not commutative: $4 \neq -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-M-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • MAIN • Q5/5

SET 18 • 7.NS.A.1.d

BACKWARD 1

Apply properties to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

Use the distributive property to rewrite 3(x + 5).

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q1/5

697

QUESTION ID

S18-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3x + 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q1/5

Combine like terms.

4y + 2 + 6y

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q2/5

698

QUESTION ID

S18-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10y + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q2/5

Which expression is equivalent to 2(4 + n)?

A. 4 + 2n

B. 6 + n

C. 8 + 2n

D. 8 + n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q3/5

699

QUESTION ID

S18-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 8 + 2n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q3/5

True or false?

$5 + (a + 7) = (5 + 7) + a$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q4/5

700

QUESTION ID

S18-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q4/5

True or false?

$6(x + 2) = 6x + 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q5/5

701

QUESTION ID

S18-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • 7.NS.A.1.d

BACKWARD 2

Add multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Add.

14.62 + 7.9

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q1/5

703

QUESTION ID

S18-B2-Q1

TYPE

Short answer

CORRECT ANSWER

22.52

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q1/5

Add.

0.75 + 3.6 + 1.25

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q2/5

704

QUESTION ID

S18-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q2/5

Add.

9.08 + 2.5

A. 34.08

B. 11.58

C. 9.33

D. 11.13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q3/5

705

QUESTION ID

S18-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 11.58

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q3/5

True or false?

$2.25 + 3.75 = 6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q4/5

706

QUESTION ID

S18-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q4/5

True or false?

$4.5 + 0.55 = 5.5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q5/5

707

QUESTION ID

S18-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • 7.NS.A.1.d

BACKWARD 3

Reorder and regroup addends to add within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add by first adding the two numbers that make a ten.

28 + 17 + 2

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 3 • Q1/5

709

QUESTION ID

S18-B3-Q1

TYPE

Short answer

CORRECT ANSWER

47

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 3 • Q1/5

Add by first adding the two numbers that make a ten.

36 + 25 + 4

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 3 • Q2/5

710

QUESTION ID

S18-B3-Q2

TYPE

Short answer

CORRECT ANSWER

65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 3 • Q2/5

Which grouping makes 46 + 19 + 4 easier to add?

A. (46 + 19) − 4

B. (46 + 4) + 19

C. 46 + 19 − 4

D. (46 − 4) + 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 3 • Q3/5

QUESTION ID

S18-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (46 + 4) + 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 3 • Q3/5

True or false?

$27 + 18 + 3 = 27 + 3 + 18$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 3 • Q4/5

712

QUESTION ID

S18-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 3 • Q4/5

True or false?

$(35 + 15) + 9 = 35 + (15 + 8)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 3 • Q5/5

713

QUESTION ID

S18-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 18 • BACKWARD 3 • Q5/5

SET 18 • 7.NS.A.1.d

FORWARD 1

Solve equations by collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$-2.5x + 4 + 1.5x = -3$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q1/5

715

QUESTION ID

S18-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q1/5

Solve for x.

$\frac{1}{3}x - 5 + \frac{2}{3}x = 9$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q2/5

716

QUESTION ID

S18-F1-Q2

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q2/5

Solve for x.

$3x - 7 + x = 2x + 5$

A. $x = 12$

B. $x = 2$

C. $x = -1$

D. $x = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q3/5

717

QUESTION ID

S18-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q3/5

True or false?

$x = -4$ is the solution of $5x + 3 - 2x = -9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q4/5

718

QUESTION ID

S18-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q4/5

True or false?

$x = 2$ is the solution of $-x + 6 + 3x = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q5/5

719

QUESTION ID

S18-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 1 • Q5/5

SET 18 • 7.NS.A.1.d

FORWARD 2

Use the structure of an expression to rewrite it • High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $x^2 - 9$ as a product of two factors.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q1/5

721

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 3)(x − 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q1

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q1/5

Factor out the greatest common factor.

$4x^2 + 12x$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q2/5

722

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4x(x + 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q2

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q2/5

Which expression is equivalent to $x^2 - 25$?

A. $(x - 5)(x + 5)$

B. $(x + 5)^2$

C. $(x - 5)^2$

D. $(x - 25)(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q3/5

723

QUESTION ID

S18-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $(x - 5)(x + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q3

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q3/5

True or false?

$x^2 + 6x + 9 = (x + 3)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q4/5

724

QUESTION ID

S18-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q4

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q4/5

True or false?

$x^2 - 16 = (x - 4)^2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q5/5

QUESTION ID

S18-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 - 16 = (x - 4)(x + 4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-F2-Q5

High School (Grade 9 in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 18 • FORWARD 2 • Q5/5

SET 19 • 7.NS.A.2.a • THE NUMBER SYSTEM

Multiply rational numbers using the rules for signs

MAIN	Grade 7 main question	**7.NS.A.2.a**
BACKWARD 1	Multiply fractions	**5.NF.B.4.a**
BACKWARD 2	Multiply decimals	**5.NBT.B.7**
FORWARD 1	Apply the properties of integer exponents with negative bases	**8.EE.A.1**
FORWARD 2	Multiply polynomials	**HSA-APR.A.1**

SET 19 • 7.NS.A.2.a

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.a

Multiply.

-6×7

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q1/5

728

QUESTION ID

S19-M-Q1

TYPE

Short answer

CORRECT ANSWER

−42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q1/5

Multiply.

$-2.5 \times (-4)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q2/5

729

QUESTION ID

S19-M-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q2/5

Multiply.

$-\frac{2}{3} \times \frac{9}{10}$

A. $-\frac{3}{5}$

B. $\frac{3}{5}$

C. $-\frac{27}{20}$

D. $-\frac{11}{13}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q3/5

730

QUESTION ID

S19-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q3/5

True or false?

$(-1)(-1) = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q4/5

731

QUESTION ID

S19-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q4/5

True or false?

The product of three negative numbers is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q5/5

732

QUESTION ID

S19-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 19 • MAIN • Q5/5

SET 19 • 7.NS.A.2.a

BACKWARD 1

Multiply fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$\frac{3}{5} \times \frac{10}{9}$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q1/5

734

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q1/5

Multiply.

$\frac{4}{7} \times 14$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q2/5

735

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q2/5

Multiply.

$\frac{5}{6} \times \frac{3}{4}$

A. $\frac{5}{8}$

B. $\frac{3}{8}$

C. $\frac{15}{10}$

D. $\frac{8}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q3/5

736

QUESTION ID

S19-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q4/5

737

QUESTION ID

S19-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q4/5

True or false?

$\frac{3}{4} \times 8 = 24$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q5/5

738

QUESTION ID

S19-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 19 • BACKWARD 1 • Q5/5

SET 19 • 7.NS.A.2.a

BACKWARD 2

Multiply decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Multiply.

2.4 × 3.5

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q1/5

740

QUESTION ID

S19-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q1/5

Multiply.

0.6 × 0.7

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q2/5

741

QUESTION ID

S19-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0.42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q2/5

Multiply.

1.25 × 8

A. 100

B. 1

C. 9.25

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q3/5

742

QUESTION ID

S19-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q3/5

True or false?

$0.3 \times 0.3 = 0.09$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q4/5

743

QUESTION ID

S19-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q4/5

True or false?

$4.2 \times 10 = 4.20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q5/5

744

QUESTION ID

S19-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • 7.NS.A.2.a

FORWARD 1

Apply the properties of integer exponents with negative bases • Grade 8 • Expressions & Equations • 8.EE.A.1

Write as a single power of −2. Then find its value.

$(-2)^5 \div (-2)^3$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q1/5

746

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(-2)^2 = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q1/5

Write as a single power of -3. Then find its value.

$(-3)^4 \times (-3)^{-1}$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q2/5

747

QUESTION ID

S19-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(-3)^3 = -27$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q2/5

Which expression is equal to $\frac{1}{16}$?

A. -4^{-2}

B. $(-4)^{-1} \times 4$

C. $(-4)^{2}$

D. $(-4)^{-2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q3/5

748

QUESTION ID

S19-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $(-4)^{-2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q3/5

True or false?

$((-5)^2)^3 = (-5)^6 = 15{,}625$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q4/5

749

QUESTION ID

S19-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q4/5

True or false?

$(-7)^0 = -1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q5/5

750

QUESTION ID

S19-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Any nonzero number to the 0 power is 1, so $(-7)^0 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 19 • FORWARD 1 • Q5/5

SET 19 • 7.NS.A.2.a

FORWARD 2

Multiply polynomials • High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply.

$-3x(2x - 5)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q1

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q1/5

752

QUESTION ID

S19-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$-6x^2 + 15x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F2-Q1**

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q1/5

Multiply.

$(x - 4)(x + 6)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q2

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q2/5

753

QUESTION ID

S19-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + 2x - 24$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F2-Q2**

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$(-2x)(-7x)$

A. $14x$

B. $14x^2$

C. $-14x^2$

D. $-9x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q3/5

754

QUESTION ID

S19-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $14x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q3

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q3/5

True or false?

$(x - 3)(x - 3) = x^2 - 6x + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q4/5

755

QUESTION ID

S19-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q4

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q4/5

True or false?

$-4(x - 2) = -4x - 8$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q5/5

QUESTION ID

S19-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-4(x - 2) = -4x + 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S19-F2-Q5

High School (Grade 9 in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • FORWARD 2 • Q5/5

SET 20 • 7.NS.A.2.a • THE NUMBER SYSTEM

Interpret products of rational numbers in real-world situations

MAIN	Grade 7 main question	**7.NS.A.2.a**
BACKWARD 1	Multiply fractions and mixed numbers in word problems	**5.NF.B.6**
BACKWARD 2	Explain the meaning of signed numbers and 0	**6.NS.C.5**
FORWARD 1	Solve equations that compare two quantities changing at negative…	**8.EE.C.7.b**
FORWARD 2	Create a one-variable equation for a situation with a negative rate	**HSA-CED.A.1**

SET 20 • 7.NS.A.2.a

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.a

A stock's price drops $1.75 each day for 6 days.

What is the total change in price?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q1/5

759

QUESTION ID

S20-M-Q1

TYPE

Short answer

CORRECT ANSWER

−$10.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q1/5

A diver descends 2.5 feet each second.

What is the change in her position after 8 seconds?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q2/5

760

QUESTION ID

S20-M-Q2

TYPE

Short answer

CORRECT ANSWER

−20 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q2/5

A tank loses 3 gallons each minute.

Which product gives the change in the amount of water after 12 minutes?

A. $12 \times 3 = 36$

B. $12 \times (-3) = -36$

C. $12 + (-3) = 9$

D. $-12 \times (-3) = 36$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q3/5

761

QUESTION ID

S20-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $12 \times (-3) = -36$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q3/5

True or false?

The temperature falls 4°F each hour for 5 hours. The total change is −20°F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q4/5

762

QUESTION ID

S20-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q4/5

True or false?

An account balance changes by −$15 each week. The change over 3 weeks is $45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q5/5

763

QUESTION ID

S20-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The change is −$45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 20 • MAIN • Q5/5

SET 20 • 7.NS.A.2.a

BACKWARD 1

Multiply fractions and mixed numbers in word problems • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A recipe uses $\frac{3}{4}$ cup of oats for each batch.

How many cups of oats are needed for 6 batches?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q1/5

765

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4 1/2

GRADING NOTE 9/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q1/5

A path is $2\frac{1}{2}$ miles long. Sam walks $\frac{2}{5}$ of it.

How far does he walk?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q2/5

766

QUESTION ID

S20-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1 mile

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q2/5

A board is $\frac{7}{8}$ yard long. Ty cuts a piece that is $\frac{1}{2}$ as long as the board.

How long is the piece?

A. $1\frac{3}{8}$ yards

B. $\frac{7}{4}$ yards

C. $\frac{8}{14}$ yard

D. $\frac{7}{16}$ yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q3/5

767

QUESTION ID

S20-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7/16 yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q3/5

True or false?

$\frac{2}{3}$ of a 12-foot rope is 8 feet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q4/5

768

QUESTION ID

S20-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q4/5

True or false?

4 bags each hold $\frac{2}{3}$ pound of nuts. Together they hold 3 pounds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q5/5

769

QUESTION ID

S20-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They hold 8/3 = 2 2/3 pounds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.6 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • 7.NS.A.2.a

BACKWARD 2

Explain the meaning of signed numbers and 0 • Grade 6 • The Number System • 6.NS.C.5

In a game, a score of 0 means no points won or lost.

What does a score of −12 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q1/5

771

QUESTION ID

S20-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12 points were lost (12 points below 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-B2-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q1/5

Sea level is 0 meters.

Which elevation is 30 meters below sea level?

A. −30 m

B. 30 m

C. 0 m

D. −3 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q2/5

772

QUESTION ID

S20-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. −30 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q2/5

What does a change of −$8 in a bank account mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q3/5

773

QUESTION ID

S20-B2-Q3

TYPE

Short answer

CORRECT ANSWER

$8 was taken out of the account.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q3/5

True or false?

A temperature change of −5° means it got colder.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q4/5

774

QUESTION ID

S20-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q4/5

True or false?

On a horizontal number line, positive numbers are to the left of 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q5/5

775

QUESTION ID

S20-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • 7.NS.A.2.a

FORWARD 1

Solve equations that compare two quantities changing at negative rates • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Diver A starts at −10 m and rises 0.5 m each second.
Diver B starts at −22 m and rises 1.25 m each second.

Solve −10 + 0.5t = −22 + 1.25t to find when they are at the same depth.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q1/5

777

QUESTION ID

S20-F1-Q1

TYPE

Short answer

CORRECT ANSWER

16 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q1/5

Tank A has 50 liters and drains 2.5 liters each minute. Tank B has 20 liters and drains 1 liter each minute.

Solve $50 - 2.5t = 20 - t$ to find when they hold the same amount.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q2/5

778

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

20 minutes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q2/5

Solve for x.

$-3(x + 4) = 2(x - 1) - 5$

A. $x = 1$

B. $x = 3$

C. $x = -1$

D. $x = -3.4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q3/5

779

QUESTION ID

S20-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = −1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q3/5

True or false?

t = 8 is the solution of 300 − 12.5t = 180 + 2.5t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q4/5

780

QUESTION ID

S20-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q4/5

True or false?

$x = -2$ is the solution of $-1.5x + 4 = 2(x + 5)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q5/5

781

QUESTION ID

S20-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = -12/7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F1-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 20 • FORWARD 1 • Q5/5

SET 20 • 7.NS.A.2.a

FORWARD 2

Create a one-variable equation for a situation with a negative rate • High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1

A phone battery is at 90% of its full charge. It drops by 6 percentage points of its full capacity each hour.

Write and solve an equation to find when it reaches 30%.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q1/5

783

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The equation 90 − 6h = 30 and the answer h = 10 hours

GRADING NOTE Both parts are required: a valid equation (for example 90 − 6h = 30) and the correct answer, 10 hours. Any valid equation is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q1

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q1/5

A submarine at −50 m dives 20 m each minute.

Write and solve an equation to find when it reaches −170 m.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q2/5

784

QUESTION ID

S20-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The equation −50 − 20t = −170 and the answer t = 6 minutes

GRADING NOTE Both parts are required: a valid equation (for example −50 − 20t = −170) and the correct answer, 6 minutes. Any valid equation is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q2

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q2/5

A store's profit was −$400 and goes up $125 each month.

Which equation finds the month m when the profit is $600?

A. −400 + 125m = 600

B. 125 − 400m = 600

C. −400 − 125m = 600

D. 400 + 125m = 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q3/5

785

QUESTION ID

S20-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A.

−400 + 125m = 600

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q3

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q3/5

True or false?

A 72°F room cools 4°F each hour. The equation $72 - 4t = 40$ finds when the room is 40°F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q4/5

QUESTION ID

S20-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q4

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q4/5

True or false?

The solution of $72 - 4t = 40$ is $t = 28$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q5/5

787

QUESTION ID

S20-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $t = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S20-F2-Q5

High School (Grade 9 in this bank) • Algebra: Creating Equations • HSA-CED.A.1 • Set 20 • FORWARD 2 • Q5/5

SET 21 • 7.NS.A.2.b • THE NUMBER SYSTEM

Divide integers and explain the sign of a quotient

MAIN	Grade 7 main question	**7.NS.A.2.b**
BACKWARD 1	Divide multi-digit whole numbers	**6.NS.B.2**
BACKWARD 2	Divide fractions	**6.NS.A.1**
FORWARD 1	Classify numbers as rational or irrational	**8.NS.A.1**
FORWARD 2	Decide whether a product is rational or irrational	**HSN-RN.B.3**

SET 21 • 7.NS.A.2.b

MAIN

Grade 7 main question • Grade 7 • The Number System • 7.NS.A.2.b

Divide.

$-48 \div 6$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q1/5

790

QUESTION ID

S21-M-Q1

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q1/5

Divide.

$-63 \div (-9)$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q2/5

791

QUESTION ID

S21-M-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q2/5

Which fraction is equal to $-\frac{3}{4}$?

A. $\frac{-3}{4}$

B. $\frac{4}{-3}$

C. $\frac{-3}{-4}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q3/5

792

QUESTION ID

S21-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (−3)/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q3/5

True or false?

$$\frac{-10}{2} = \frac{10}{-2} = -5$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q4/5

793

QUESTION ID

S21-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q4/5

True or false?

$-7 \div 0 = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q5/5

794

QUESTION ID

S21-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Division by 0 is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-M-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 21 • MAIN • Q5/5

SET 21 • 7.NS.A.2.b

BACKWARD 1

Divide multi-digit whole numbers • Grade 6 • The Number System • 6.NS.B.2

ANSWER KEY Reduced copy of the question page, with its answer

Divide.

1,872 ÷ 24

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q1/5

796

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

78

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q1/5

Divide.

3,915 ÷ 45

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q2/5

797

QUESTION ID

S21-B1-Q2

TYPE

Short answer

CORRECT ANSWER

87

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q2/5

Divide.

2,436 ÷ 12

A. 203

B. 2,030

C. 23

D. 230

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q3/5

798

QUESTION ID

S21-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 203

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S21-B1-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q3/5

True or false?

5,040 ÷ 35 = 144

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q4/5

799

QUESTION ID

S21-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q4/5

True or false?

1,950 ÷ 15 = 13

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q5/5

QUESTION ID

S21-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 21 • BACKWARD 1 • Q5/5